Yang Hui, Xiangjie Jiuzhang Suanfa, parts 1-3

English Translation

Modern LaTeX Editions of Public-Domain Mathematics Manuscripts

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Detailed Analysis of the Mathematical Methods in the Nine Chapters

詳解九章算法

(Part I — 上)

Bilingual Edition / 中英雙語版

By **Yang Hui** 楊輝
Southern Song Dynasty (c. 1238–1298 CE)

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*The "Method Source for Square Root Extraction" (*開方作法本源圖*)*

Left Column: Original Chinese text / Right Column: English translation

Typeset in XeLaTeX with Noto Serif CJK SC
May 27, 2026

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1 Introduction / 引言

《詳解九章算法》乃南宋楊輝所著之重要數學著作。成書於一二六一年，詳解古代《九章算術》之各題算法。

Xiangjie Jiuzhang Suanfa (詳解九章算法, "Detailed Analysis of the Mathematical Methods in the Nine Chapters") is one of the most important mathematical works of the Southern Song Dynasty, written by the mathematician **Yang Hui** 楊輝 (c. 1238–1298 CE). Completed in 1261, this work provides detailed explanations and commentaries on the classic *Jiuzhang Suanshu* (Nine Chapters on the Mathematical Art).

1.1 Historical Context / 歷史背景

原《九章算術》成書於漢代（約公元前二百年至公元二百年），凡二百四十六題，分九章。至宋代，原文因靖康之變（一一二七年）而散佚，傳抄訛誤，題隱法晦，學者難以入門。

The original *Jiuzhang Suanshu* dates back to the Han Dynasty (c. 200 BCE–200 CE) and contains 246 mathematical problems organized into nine chapters. By the Song Dynasty, the original text had become difficult to understand due to the loss of manuscripts during the *Jingkang Incident* (靖康之變, 1127 CE), corruption of transmitted texts through repeated copying, and obscure problem statements with difficult methods.

輝嘗聞學者，謂九章題問頗隱，法理難明，不得其門而入。於是以答參問，用草考法，因法推類，然後知斯文非古之全經也。

"I have heard scholars say that the problems in the Nine Chapters are rather obscure, and the methods difficult to understand, so one cannot find the door to enter. Therefore, using the answers to examine the questions, employing the worked solutions to investigate the methods, and following the methods to extend by analogy—only then do I know that this text is not the complete classic of antiquity."

1.2 Structure of the Work / 全書結構

全書凡十二章：一、乘除（四十一題）；二、方田（三十八題）；三、粟米（四十六題）；四、衰分（二十題）；五、少廣（二十四題）；六、商功（二十八題）；七、均輸（二十八題）；八、盈不足（二十題）；九、方程（十八題）；十、勾股（三十八題）；十一、纂類。

The original work comprised **12 chapters**: (1) Multiplication and Division — 41 problems; (2) Field Measurement — 38 problems; (3) Grain Conversion — 46 problems; (4) Proportional Distribution — 20 problems; (5) Short Width / Root Extraction — 24 problems; (6) Commercial Calculations / Engineering — 28 problems; (7) Equitable Distribution — 28 problems; (8) Excess and Deficit — 20 problems; (9) Systems of Linear Equations — 18 problems; (10) Right Triangles — 38 problems; (11) Classification / Compilation.

1.3 Editorial Method / 編纂之法

楊輝解題之法，分為四部：問（設題）、答（給出數值答案）、術（說明一般算法）、草（詳細逐步演算）。此四部結構，為中國數學著述之典範。

Yang Hui's approach to explaining the Nine Chapters was systematic: **Problem (問)** states the mathematical problem; **Answer (答)** gives the numerical answer; **Method (術)** states the general algorithm; **Worked Solution (草)** provides a detailed step-by-step calculation. This four-part structure became a model for Chinese mathematical exposition.

1.4 The Method Source for Square Root Extraction / 開方作法本源

楊輝所錄之開方作法本源圖（今稱楊輝三角或賈憲三角），源出北宋賈憲（約一〇一〇至一〇七〇年）之《釋鎖算書》。其圖列二項式係數至六次冪，用於開方運算。

One of Yang Hui's most celebrated contributions is preserving what we now call **Yang Hui's Triangle** (also known as Pascal's Triangle). In his work, he attributes this to the earlier mathematician **Jia Xian** (c. 1010–1070 CE), who called it the "Method Source for Square Root Extraction" (開方作法本源圖). The triangle provides binomial coefficients used in root extraction. Yang Hui wrote: 出釋鎖算書，賈憲用此術 ("This comes from the *Shisuo Suan Shu*; Jia Xian used this method").

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2 Original Preface / 原序

黃帝九章古序云：國家嘗設算科取士，選九章以為算經之首，蓋猶儒者之六經，醫家之難素，兵法之孫子歟。

"The ancient preface to the Yellow Emperor's Nine Chapters states: The state once established an examination in mathematics for selecting officials, choosing the Nine Chapters as the foremost mathematical classic—this is like the Six Classics for Confucian scholars, the *Nanjing* and *Suwen* for physicians, or the *Art of War* for military strategists."

昔聖宋紹興戊辰，算士榮棨謂靖康以來，罕有舊本，間有存者，狃於末習，向獲善本，得其全經，復起於學。以魏景元元年劉徽等、唐朝議大夫、行太史令上輕車都尉李淳風等注釋，聖宋右班直賈憲撰草。

"In the past, during the Wuchen year of the Shaoxing reign of our Sacred Song Dynasty, the mathematician Rong Qi said that since the Jingkang period, old copies have been rare, and those occasionally surviving have been corrupted by inferior practices. Having obtained a good copy and recovered the complete classic, study was revived. The annotations were by Liu Hui et al. (Jingyuan first year of Wei) and by Li Chunfeng et al. (Senior Compiler of Tang Dynasty, Acting Grand Astrologer), while the detailed solutions were composed by Jia Xian of the Song Dynasty."

輝嘗聞學者，謂九章題問頗隱，法理難明，不得其門而入。於是以答參問，用草考法，因法推類，然後知斯文非古之全經也。將後賢補贅之文，修前代已廢之法，刪立題術，又篡法問，詳著於後，儻得賢者，改而正諸，是所願也。

"I have heard scholars say that the problems in the Nine Chapters are rather obscure, and the principles difficult to understand, so one cannot find the door to enter. Therefore, using the answers to examine the questions, employing the worked solutions to investigate the methods, and following the methods to extend by analogy—only then do I know that this text is not the complete classic of antiquity. Taking the supplementary texts added by later worthies and repairing the methods abandoned by earlier generations, I have reorganized the problem-methods and compiled the method-questions, recording them in detail below. If worthy scholars would correct and improve them, that is my wish."

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3 Chapter 1: Multiplication and Division / 乘除

第一章論乘除之基本運算，凡四十一題，多取方田章及粟米之題。

The first chapter covers fundamental operations of multiplication and division, with 41 problems drawn mostly from the "Field Measurement" (方田) chapter and grain conversion problems.

3.1 Rectangular Field Method / 直田法

術：以廣乘從得積。以畝法（一畝二百四十平方步）除之。

Method: Multiply the width (廣) by the length (從) to obtain the area (積). Divide by the unit conversion (畝法, 1 mu = 240 square bu).

PROBLEM 1

廣十五步，從十六步，問田幾何？答曰：一畝。 A field has width 15 bu and length 16 bu. What is the area? **Answer**: 1 mu.

SOLUTION (草曰):

廣十五步，從十六步。積 = $15 \times 16 = 240$ 平方步。一畝 = 二百四十平方步，故為**一畝**。

Width = 15 bu, Length = 16 bu. Area = $15 \times 16 = 240$ *square bu. Since 1 mu = 240 square bu, the area is* ***1 mu****.*

PROBLEM 2

廣十二步，從十四步，問田幾何？答曰：一百六十八步。 A field has width 12 bu and length 14 bu. What is the area in square bu? **Answer**: 168 square bu.

SOLUTION (草曰):

積 = $12 \times 14 = \mathbf{168}$ 平方步。

Area = $12 \times 14 = \mathbf{168}$ *square bu.*

3.2 Square Field Method / 方田法

術：方田之積，以里自乘，復以三百七十五畝乘之。

Method: Square the side length to obtain the area in square li. Multiply by 375 mu per square li for conversion.

PROBLEM 3

方田一里，問為田幾何？答曰：三頃七十五畝。 A square field has side 1 li. What is its area? **Answer**: 3 qing 75 mu.

SOLUTION (草曰):

積 = $1 \times 1 = 1$ 平方里。$1 \times 375 =$ **375 畝** = **三頃七十五畝**。

Area = $1 \times 1 = 1$ *square li.* $1 \times 375 =$ ***375 mu*** = ***3 qing 75 mu****.*

3.3 Fraction Multiplication / 乘分

術：分母乘全步，并子以為實。分母相乘為法。以法除實。

Method: For fractions with whole numbers: multiply each whole number by its denominator and add the numerator. Multiply these results to form the dividend (實). Multiply the denominators to form the divisor (法). Divide the dividend by the divisor.

PROBLEM Fraction Multiplication

田廣七步、四分步之三，從十五步、九分步之五，問：為田幾何？答曰：一百二十步，九分步之五。 A field has width $7\,\frac{3}{4}$ bu and length $15\,\frac{5}{9}$ bu. What is the area? **Answer**: $120\,\frac{5}{9}$ square bu.

SOLUTION (草曰):

廣：$7\,\frac{3}{4} = \frac{31}{4}$ 步。從：$15\,\frac{5}{9} = \frac{140}{9}$ 步。
積 = $\frac{31}{4} \times \frac{140}{9} = \frac{4340}{36} = 120\,\frac{5}{9}$ 平方步。

Width: $7\,\frac{3}{4} = \frac{31}{4}$ *bu. Length:* $15\,\frac{5}{9} = \frac{140}{9}$ *bu.*
Area = $\frac{31}{4} \times \frac{140}{9} = \frac{4340}{36} = 120\,\frac{5}{9}$ *square bu.*

3.4 Division with Fractions / 除分

術：以人數為法，錢數為實。有分者通之。實如法而一。

Method: Use the number of people as the divisor (法) and the amount of money as the dividend (實). If there are fractions, convert them. Divide the dividend by the divisor.

Problem Division with Fractions

七人，均八錢三分錢之一，問人得幾何？答曰：人得一錢二十一分錢之四。 Seven people share $8\,\frac{1}{3}$ qian equally. How much does each person receive? **Answer**: $1\,\frac{4}{21}$ qian each.

Solution (草曰):

總數：$8\,\frac{1}{3} = \frac{25}{3}$ 錢。
每人得：$\frac{25}{3} \div 7 = \frac{25}{21} = 1\,\frac{4}{21}$ 錢。

Total: $8\,\frac{1}{3} = \frac{25}{3}$ *qian.*
Each receives: $\frac{25}{3} \div 7 = \frac{25}{21} = 1\,\frac{4}{21}$ *qian.*

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4 Chapter 2: Field Measurement / 方田

方田章專論面積之計算與分數之運算。凡三十八題，涉及各種田形及分數四則運算。

The Field Measurement chapter focuses on area calculations and fraction operations. It originally contained 38 problems covering various field shapes and operations with fractions.

4.1 Reduction of Fractions / 約分

術： 可半者半之。不可半者，副置分母子之數，以少減多，更相減損，求其等也。以等數約之。（即輾轉相除之歐幾里得算法。）

Method: If both numerator and denominator are even, halve them. If not, place the numerator and denominator separately and repeatedly subtract the smaller from the larger (Euclidean algorithm) until the greatest common divisor is found. Divide both by this common measure.

Problem Reduction

十八分之十二。問：約之得幾何？答曰：三分之二。 Reduce $\frac{12}{18}$. **Answer:** $\frac{2}{3}$.

Solution (草曰):

十二與十八之等數：$18 - 12 = 6$，$12 - 6 = 6$，$6 - 6 = 0$。等數 = 6。
$\frac{12 \div 6}{18 \div 6} =$

GCD of 12 and 18: $18 - 12 = 6, 12 - 6 = 6, 6 - 6 = 0$. *GCD = 6.*
$\frac{12 \div 6}{18 \div 6} =$

Problem

九十一分之四十九。問：約之得幾何？答曰：十三分之七。 Reduce $\frac{49}{91}$. **Answer:** $\frac{7}{13}$.

Solution (草曰):

輾轉相除：$91 - 49 = 42$，$49 - 42 = 7$，$42 - 7 \times 6 = 0$。等數 = 7。
$\frac{49 \div 7}{91 \div 7} =$

Using the Euclidean algorithm: $91 - 49 = 42, 49 - 42 = 7, 42 - 7 \times 6 = 0$. *GCD = 7.*
$\frac{49 \div 7}{91 \div 7} =$

4.2 Addition of Fractions / 合分

術： 母互乘子，并以為實。母相乘為法。實如法而一。母同者直相從之。

Method: Cross-multiply the denominators with the numerators and add to form the dividend. Multiply the denominators to form the divisor. Divide. If the denominators are the same, simply add the numerators.

Problem

二分之一、三分之二、四分之三、五分之四，合之得幾何？答曰：得二，餘六十分之四十三。 Add $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5}$. **Answer:** $2\,\frac{43}{60}$.

Solution (草曰):

互乘：$1 \times 3 \times 4 \times 5 + 2 \times 2 \times 4 \times 5 + 3 \times 2 \times 3 \times 5 + 4 \times 2 \times 3 \times 4 = 60 + 80 + 90 + 96 = 326$。
法：$2 \times 3 \times 4 \times 5 = 120$。
$\frac{326}{120} = 2\,\frac{86}{120} = 2\,\frac{43}{60}$。

Cross-multiply: $1 \times 3 \times 4 \times 5 + 2 \times 2 \times 4 \times 5 + 3 \times 2 \times 3 \times 5 + 4 \times 2 \times 3 \times 4 = 60 + 80 + 90 + 96 = 326$.
Divisor: $2 \times 3 \times 4 \times 5 = 120$.
$\frac{326}{120} = 2\,\frac{86}{120} = 2\,\frac{43}{60}$.

4.3 Subtraction and Comparison of Fractions / 課分

術： 母互乘子，以少減多，餘為實。母相乘為法。實如法而一。即兩分數之差也。

Method: Cross-multiply numerators, subtract the smaller from the larger. The remainder is the dividend, the product of denominators is the divisor. This gives the difference (also called "the excess").

Problem

八分之五，比二十五分之十六。問：孰多多幾何？答曰：二十五分之十六，多多二百分之三。Compare $\frac{5}{8}$ and $\frac{16}{25}$. Which is larger and by how much? **Answer**: $\frac{16}{25}$ is larger by $\frac{3}{200}$.

Solution (草曰):

$\frac{16}{25} - \frac{5}{8} = \frac{16\times8-5\times25}{25\times8} = \frac{128-125}{200} = \frac{3}{200}$。
故 $\frac{16}{25}$ 大 $3_{\overline{200}}$
$\frac{16}{25} - \frac{5}{8} = \frac{16\times8-5\times25}{25\times8} = \frac{128-125}{200} = \frac{3}{200}$.
So $\frac{16}{25}$ is larger by

4.4 Equalizing Fractions / 平分

術：母互乘子，各為列實。并諸列實為平實。母相乘為法。以列數乘未并者，各為列實。以平實減列實，餘為減數；以列實減平實，餘為益數。減多益少，使各平於平實。

Method: Cross-multiply numerators. Sum all numerators for the "level dividend." Multiply all denominators for the divisor. Multiply the number of terms by each uncrossed numerator for the column dividends. Subtract the level dividend from each column dividend to find what must be subtracted; subtract each column dividend from the level dividend to find what must be added. Combine the subtractions to benefit the smaller values. Divide the level dividend by the divisor to find the equalized value.

Problem

二分之一、三分之二、四分之三，減多益少幾何？而平？答曰：減四分之三，求之者四，減三分之二，求之者一，益二分之一，求之者五，各平於三十六分之二十三。 Given $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, redistribute by subtracting from larger and adding to smaller to make them equal. **Answer**: Subtract $\frac{4}{36}$ from $\frac{3}{4}$; subtract $\frac{1}{36}$ from $\frac{2}{3}$; add $\frac{5}{36}$ to $\frac{1}{2}$. All equalized to $\frac{23}{36}$.

Solution (草曰):

平值 = $\frac{23}{36}$。
$\frac{3}{4} = \frac{27}{36}$：減 $\frac{4}{36}$。
$\frac{2}{3} = \frac{24}{36}$：減 $\frac{1}{36}$。
$\frac{1}{2} = \frac{18}{36}$：益 $\frac{5}{36}$。
皆平於 $23_{\overline{36}}$

Level value = $\frac{23}{36}$.
$\frac{3}{4} = \frac{27}{36}$: *subtract* $\frac{4}{36}$.
$\frac{2}{3} = \frac{24}{36}$: *subtract* $\frac{1}{36}$.
$\frac{1}{2} = \frac{18}{36}$: *add* $\frac{5}{36}$.
All equalized to $23_{\overline{36.}}$

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5 Chapter 3: Grain Conversion / 粟米

粟米章論各種穀物與商品之比例交換。凡四十六題，其中六題為基本乘除，三十一題為比例交換，九題為比例率。

The Grain Conversion chapter deals with proportional exchange between different types of grain and commodities. It originally contained 46 problems with 6 on basic multiplication/division, 31 on proportional exchange, and 9 on proportional rates.

5.1 Direct Exchange / 經率

術：以所買率為法，以所出錢為實。實如法得一。

Method: Use the money amount as the dividend (實) and the quantity purchased as the divisor (法). Divide the dividend by the divisor.

Problem

錢一百六十文，買瓴甓十八枚，問枚價幾何？答曰：一枚八錢九分钱之八。 With 160 qian, buy 18 roof tiles. What is the price per tile? **Answer**: $8\frac{8}{9}$ qian per tile.

Solution (草曰):

$160 \div 18 = \frac{160}{18} = \frac{80}{9} = 8\frac{8}{9}$ 文/枚。

$160 \div 18 = \frac{160}{18} = \frac{80}{9} = 8\frac{8}{9}$ *qian per tile.*

5.2 Grain Conversion Rates / 粟米換算率

《九章》立穀物換算之標準率：粟率五十；糲米率三十；糳米率二十七；御米率二十四；麥率四十五；菽率四十五。

The Nine Chapters establishes standard conversion rates between different grains: Millet (粟): rate = 50; Coarse rice (糲米): rate = 30; Fine rice (糳米): rate = 27; Polished rice (御米): rate = 24; Wheat (麥): rate = 45; Soybeans (菽): rate = 45.

術：以所有數乘所求率為實，以所有率為法，實如法而一。

Method: "What you seek multiplied by the given amount, divided by what you have." (以所有數乘所求率為實，以所有率為法，實如法而一。)

Problem

粟一斗。問為糲米幾何？答曰：六升。 Given 1 dou of millet, how much coarse rice (糲米) does it make? **Answer**: 6 sheng.

Solution (草曰):

粟率五十，糲米率三十。
1 斗 $\times \frac{30}{50} = \frac{3}{5}$ 斗 $=$ ***6* 升**。

Millet rate = 50, coarse rice rate = 30.
1 *dou* $\times \frac{30}{50} = \frac{3}{5}$ *dou* $=$ ***6 sheng***.

Problem

粟四斗五升。問為糳米幾何？答曰：二斗一升五分升之三。 Given 4 dou 5 sheng of millet, how much fine rice (糳米) does it make? **Answer**: 2 dou 1 sheng $\frac{3}{5}$.

Solution (草曰):

$4.5 \times \frac{27}{50} = \frac{9}{2} \times \frac{27}{50} = \frac{243}{100} = 2.43$ 斗 = **二斗一升五分升之三**。

$4.5 \times \frac{27}{50} = \frac{9}{2} \times \frac{27}{50} = \frac{243}{100} = 2.43$ *dou* = ***2 dou 1 sheng*** $\frac{3}{5}$.

Problem

糲米十五斗五升五分升之二。問為粟幾何？答曰：二十五斗九升。 Given 15 dou $5\frac{2}{5}$ sheng of coarse rice, how much millet was the original amount? **Answer**: 25 dou 9 sheng.

Solution (草曰):

$15\,5\frac{2}{5} = \frac{777}{50}$ 斗。
$\frac{777}{50} \times \frac{50}{30} = \frac{777}{30} = 25.9$ 斗 = **二十五斗九升**。

$15\,5\frac{2}{5} = \frac{777}{50}$ *dou.*
$\frac{777}{50} \times \frac{50}{30} = \frac{777}{30} = 25.9$ *dou* = ***25 dou 9 sheng***.

5.3 Silk and Currency Conversion / 絲錢換算

Problem

絲一斤，價三百四十五，今有七兩一十二銖。問錢，答曰：一百六十一錢，三十二分錢之二十三。 Silk costs 345 qian per jin. If one has 7 liang 12 zhu of silk, what is its value in qian? **Answer**: $161\,\frac{23}{32}$ qian.

Solution (草曰):

一斤 = 十六兩 = 三百八十四銖。七兩十二銖 = $7\times24+12=180$ 銖。
值 = $345\times\frac{180}{384}=345\times\frac{15}{32}=\frac{5175}{32}=161\,\frac{23}{32}$ 錢。

1 jin = 16 liang = 384 zhu. 7 liang 12 zhu = $7\times24+12=180$ *zhu.*
Value = $345\times\frac{180}{384}=345\times\frac{15}{32}=\frac{5175}{32}=161\,\frac{23}{32}$ *qian.*

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6 Chapter 4: Proportional Distribution / 衰分

衰分章論按給定比例分配數量之問題。凡二十題。

The Proportional Distribution chapter deals with dividing quantities in given proportions. It originally contained 20 problems.

6.1 The Method of Distribution / 衰分法

術：各置列衰，副并為法。以所分乘未并者，各自為實。實如法而一。不滿法者，以法命之。

Method: Set out the distribution rates (衰). Combine them as the divisor. Multiply what is to be distributed by each uncombined rate to form the column dividends. Divide each by the divisor. If there is a remainder, express it as a fraction over the divisor.

Problem Five Ranks Sharing Deer / 五爵分鹿

大夫、不更、簪裹、上造、公士凡五人，以爵次高下均五鹿，問各幾何？答曰：大夫一鹿，三分鹿之二；不更一鹿，三分鹿之一；簪裹一鹿；上造鹿三分之二；公士鹿三分之一。 Five nobles—a Dafu, a Bugeng, a Zanmao, a Shangzao, and a Gongshi—are to share 5 deer in proportion to their ranks (5:4:3:2:1). **Answer**: Dafu $1\frac{2}{3}$; Bugeng $1\frac{1}{3}$; Zanmao 1; Shangzao $\frac{2}{3}$; Gongshi $\frac{1}{3}$ deer.

Solution (草曰):

衰：大夫五、不更四、簪裹三、上造二、公士一。
并 $= 5 + 4 + 3 + 2 + 1 = 15$。
大夫：$5 \times \frac{5}{15} = \frac{25}{15} = 1\frac{2}{3}$ 鹿。
不更：$5 \times \frac{4}{15} = \frac{20}{15} = 1\frac{1}{3}$ 鹿。
簪裹：$5 \times \frac{3}{15} = 1$ 鹿。
上造：$5 \times \frac{2}{15} = \frac{2}{3}$ 鹿。
公士：$5 \times \frac{1}{15} = \frac{1}{3}$ 鹿。

Rates: Dafu 5, Bugeng 4, Zanmao 3, Shangzao 2, Gongshi 1.
Sum $= 5 + 4 + 3 + 2 + 1 = 15$.
Dafu: $5 \times \frac{5}{15} = \frac{25}{15} = 1\frac{2}{3}$ *deer.*
Bugeng: $5 \times \frac{4}{15} = \frac{20}{15} = 1\frac{1}{3}$ *deer.*
Zanmao: $5 \times \frac{3}{15} = 1$ *deer.*
Shangzao: $5 \times \frac{2}{15} = \frac{2}{3}$ *deer.*
Gongshi: $5 \times \frac{1}{15} = \frac{1}{3}$ *deer.*

6.2 The Cow, Horse, and Sheep / 牛馬羊償粟

Problem

牛、馬、羊食人苗，苗主責之粟五斗，牛食馬之半，馬食羊之半，欲衰償之？答曰：牛二斗八升，七分升之四；馬一斗四升，七分升之二；羊七升，七分升之一。 A cow, a horse, and a sheep ate someone's crops. The owner demands 5 dou of millet as compensation. The cow ate twice what the horse ate, and the horse ate twice what the sheep ate. **Answer**: Cow $2\frac{4}{7}$ dou; Horse $1\frac{2}{7}$ dou; Sheep $\frac{1}{7}$ dou.

Solution (草曰):

衰：牛四、馬二、羊一。
并 $= 4 + 2 + 1 = 7$。
牛：$5 \times \frac{4}{7} = \frac{20}{7} = 2\frac{6}{7}$ 斗 = **二斗八升** $\frac{4}{7}$。
馬：$5 \times \frac{2}{7} = \frac{10}{7} = 1\frac{3}{7}$ 斗 = **一斗四升** $\frac{2}{7}$。
羊：$5 \times \frac{1}{7} = \frac{5}{7}$ 斗 = **七升** $\frac{1}{7}$。

The rates are: Cow 4, Horse 2, Sheep 1.
Sum $= 4 + 2 + 1 = 7$.
Cow: $5 \times \frac{4}{7} = \frac{20}{7} = 2\frac{6}{7}$ *dou =* ***2 dou 8 sheng*** $\frac{4}{7}$.
Horse: $5 \times \frac{2}{7} = \frac{10}{7} = 1\frac{3}{7}$ *dou =* ***1 dou 4 sheng*** $\frac{2}{7}$.
Sheep: $5 \times \frac{1}{7} = \frac{5}{7}$ *dou =* ***7 sheng*** $\frac{1}{7}$.

6.3 The Woman Skilled at Weaving / 女子善織

Problem

女子善織，日自倍，五日織五尺。問日織數。答曰：初日一寸，三十一分寸之十九；二日三寸，三十一分寸之七；三日六寸，三十一分寸之十四；四日一尺二寸，三十一分寸之二十八；五日二尺四寸，三十一分寸之二十五。 A woman skilled at weaving doubles her output each day. In 5 days she weaves 5 chi. **Answer**: Day 1: $\frac{50}{31}$ cun; Day 2: $\frac{100}{31}$ cun; Day 3: $\frac{200}{31}$ cun; Day 4: $\frac{400}{31}$ cun; Day 5: $\frac{800}{31}$ cun.

Solution (草曰):

設首日織 x，則總數 $= x + 2x + 4x + 8x + 16x = 31x = 5$ 尺。
$x = \frac{5}{31}$ 尺 = $\frac{50}{31}$ 寸 = $1\frac{19}{31}$ 寸。
次日：$\frac{100}{31} = 3\frac{7}{31}$ 寸。
三日：$\frac{200}{31} = 6\frac{14}{31}$ 寸。
四日：$\frac{400}{31} = 12\frac{28}{31}$ 寸。
五日：$\frac{800}{31} = 25\frac{25}{31}$ 寸。

Let the first day's output be x. *Then total* $= x + 2x + 4x + 8x + 16x = 31x = 5$ *chi.*
$x = \frac{5}{31}$ *chi* $= \frac{50}{31}$ *cun* $= 1\frac{19}{31}$ *cun.*
Day 2: $\frac{100}{31} = 3\frac{7}{31}$ *cun.*
Day 3: $\frac{200}{31} = 6\frac{14}{31}$ *cun.*
Day 4: $\frac{400}{31} = 12\frac{28}{31}$ *cun.*
Day 5: $\frac{800}{31} = 25\frac{25}{31}$ *cun.*

Original Chinese (原文) | English Translation (英譯)

7 Chapter 5: Short Width / Root Extraction / 少廣

少廣章論已知矩形面積與一邊而求另一邊，涉及分數除法與開方。凡二十四題。

The Short Width chapter deals with finding an unknown dimension of a rectangle given its area and one side, which leads to problems of division with fractions and root extraction. It originally contained 24 problems.

7.1 The Short Width Method / 少廣法

術：置全步及分母子。各以分母乘其子、全步，以少減多，通分內子，令母互乘。并所有以為法。以畝積二百四十步乘之為實。實如法而一，得從。

Method: Set out the whole steps and the fraction denominators and numerators. Separately multiply the denominators by themselves, then multiply the whole steps and numerators. Divide each by its original denominator and combine. Use this as the divisor. Multiply the whole steps by their combined denominators, then multiply by the field area (mu) in square bu for the dividend. Divide the dividend by the divisor.

Problem Short Width 1

田一畝廣一步半，問從？答曰：一百六十步。 A field of 1 mu has width $1\,\frac{1}{2}$ bu. What is the length? **Answer**: 160 bu.

Solution (草曰):

廣 = $1\,\frac{1}{2} = \frac{3}{2}$ 步。積 = 一畝 = 二百四十平方步。
從 = $240 \div \frac{3}{2} = 240 \times \frac{2}{3} = \mathbf{160}$ 步。

Width = $1\,\frac{1}{2} = \frac{3}{2}$ *bu. Area = 1 mu = 240 square bu.*
Length = $240 \div \frac{3}{2} = 240 \times \frac{2}{3} = \mathbf{160}$ *bu.*

Problem

田一畝廣一步半、三分步之一。問從？答曰：一百三十步一十一分步之一十。 A field of 1 mu has width $1\,\frac{1}{2}\,\frac{1}{3}$ bu. What is the length? **Answer**: $130\,\frac{10}{11}$ bu.

Solution (草曰):

廣 = $1 + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6}$ 步。
從 = $240 \div \frac{11}{6} = 240 \times \frac{6}{11} = \frac{1440}{11} = 130\,\frac{10}{11}$ 步。

Width = $1 + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6}$ *bu.*
Length = $240 \div \frac{11}{6} = 240 \times \frac{6}{11} = \frac{1440}{11} = 130\,\frac{10}{11}$ *bu.*

Problem

田一畝廣一步半、三分步之一、四分步之一。問從答曰：一百一十五步五分步之一。 A field of 1 mu has width $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$ bu. **Answer**: $115\,\frac{1}{5}$ bu.

Solution (草曰):

廣 = $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{12+6+4+3}{12} = \frac{25}{12}$ 步。
從 = $240 \div \frac{25}{12} = \frac{2880}{25} = 115\,\frac{1}{5}$ 步。

Width = $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{12+6+4+3}{12} = \frac{25}{12}$ *bu.*
Length = $240 \div \frac{25}{12} = \frac{2880}{25} = 115\,\frac{1}{5}$ *bu.*

Problem

田一畝廣一步半、三分步之一、四分步之一、五分步之一、六分步之一、七分步之一。問從答曰：九十二步一百二十一分步之六十八。 A field of 1 mu has width $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}$ bu. **Answer**: $92\,\frac{68}{121}$ bu.

Solution (草曰):

用逆少廣術：
廣 = $\frac{420+210+140+105+84+70+60}{420} = \frac{1089}{420}$。
約之：等數三，得 $\frac{363}{140}$。
從 = $240 \times \frac{140}{363} = \frac{33600}{363} = 92\,\frac{204}{363} = 92\,\frac{68}{121}$ 步。

Using the reverse combination-of-fractions method:
Width = $\frac{420+210+140+105+84+70+60}{420} = \frac{1089}{420}$.
Simplifying: GCD(1089, 420) = 3, so $\frac{363}{140}$.
Length = $240 \times \frac{140}{363} = \frac{33600}{363} = 92\,\frac{204}{363} = 92\,\frac{68}{121}$ *bu.*

Original Chinese (原文) | English Translation (英譯)

8 The Method Source for Square Root Extraction / 開方作法本源圖

楊輝書中最著名者，為其所錄之開方作法本源圖（今稱楊輝三角）。楊輝稱此圖源出北宋賈憲（約一〇一〇至一〇七〇年）。

One of the most celebrated parts of Yang Hui's work is his recording of what we now call **Yang Hui's Triangle** (also known as Pascal's Triangle in the West). Yang Hui attributed this to Jia Xian (c. 1010–1070 CE), who called it the "Method Source for Square Root Extraction" (開方作法本源).

8.1 The Original Diagram / 原圖

楊輝釋此圖曰：
左袤乃積數，右袤乃隅算，中藏者皆廉，以廉乘商方，命實而除之。

Yang Hui presented the triangle with the following explanation: "The left diagonal consists of the accumulated numbers; the right diagonal consists of the corner calculations. Those hidden in the middle are all the 'flanks' (intermediate coefficients). Using the flanks, multiply the quotient square, and command the dividend to be divided by it."

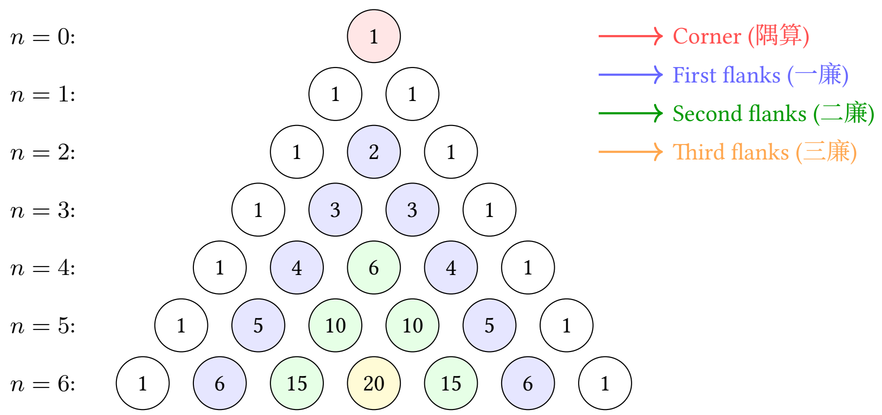

Fig. 1: Yang Hui's Triangle showing binomial coefficients up to the 6th power / 楊輝三角至六次冪

8.2 Explanation of the Structure / 結構釋義

左袤（積數）：斜邊之常數項（皆為一），即 $(x+a)^n$ 展開式之常數係數。
右袤（隅算）：最高次冪之係數（亦皆為一）。
中藏者（廉）：中間各數為廉，即各中間項之係數。

Left diagonal (左袤): The constant terms (1, 1, 1, ...) represent the "accumulated numbers" (積數)—the coefficients of the constant term in $(x+a)^n$.
Right diagonal (右袤): The coefficients of the highest power—also all 1's—represent the "corner calculations" (隅算).
Interior numbers (中藏者): The intermediate numbers are the "flanks" (廉)—the coefficients of the intermediate terms.

每行給出 $(x+a)^n$ 展開之係數：

$$\begin{aligned}(x+a)^0 &= 1\\(x+a)^1 &= x+a\\(x+a)^2 &= x^2+2xa+a^2\\(x+a)^3 &= x^3+3x^2a+3xa^2+a^3\end{aligned}$$

Each row gives the coefficients for expanding $(x+a)^n$:

$$\begin{aligned}(x+a)^0 &= 1\\(x+a)^1 &= x+a\\(x+a)^2 &= x^2+2xa+a^2\\(x+a)^3 &= x^3+3x^2a+3xa^2+a^3\end{aligned}$$

8.3 Application to Root Extraction / 開方之應用

楊輝與賈憲用此係數進行開平方與開立方。開平方之術：一、置積為實。二、商估首位。三、以三角之係數為方法、廉法、隅法。四、乘減遞推。

Yang Hui and Jia Xian used these coefficients to perform square root and cube root extraction. To extract $\sqrt{A}$: (1) Set A as the dividend (實). (2) Estimate the first digit of the root as the quotient (商). (3) Use the coefficients from the triangle to form the "method" (法), "flank" (廉), and "corner" (隅). (4) Multiply and subtract iteratively.

8.4 Jia Xian's Generating Method / 賈憲增乘方求廉法

增乘方求廉法：以一次乘二次，以二次乘三次，遞增一位，至求終。

"To seek the flanks by augmented multiplication: multiply once by twice, multiply twice by thrice, increasing by one position each time, until the end is reached."

此法由前行生成後行，相鄰兩數相加得之—即今所謂帕斯卡法則：

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

This method generates each row from the previous one by adding adjacent entries—exactly the rule we now call Pascal's rule:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Original Chinese (原文)	English Translation (英譯)

9 Chapter 6: Commercial Calculations / 商功

商功章論工程土方之體積計算。凡二十八題。

The Commercial Calculations chapter deals with volume computations for engineering projects (earthworks, construction, etc.). It originally contained 28 problems.

9.1 Volume Conversion Rates / 土方換算率

《九章》立不同土質之換算率：穿地為基準；壤（鬆土）與穿地之比為 4 : 5；堅（堅土）與穿地之比為 3 : 4。

The Nine Chapters establishes conversion rates between different soil conditions: Excavated earth (穿地): base measure; Loose earth (壤): 4 : 5 ratio with excavated; Compacted earth (堅): 3 : 4 ratio with excavated.

Problem

穿地積一萬尺。問為堅、壤各幾何？答曰：為堅七千五百尺，為壤一萬二千五百尺。 An excavation of 10,000 cubic chi. How much compacted and loose earth does it produce? **Answer**: Compacted 7,500; Loose 12,500 cubic chi.

Solution (草曰):

穿地: 壤: 堅 = 4 : 5 : 3。
壤 = $10000 \times \frac{5}{4} = \boldsymbol{12500}$ 立方尺。
堅 = $10000 \times \frac{3}{4} = \boldsymbol{7500}$ 立方尺。

Excavated : Loose : Compacted = 4 : 5 : 3.
Loose = $10000 \times \frac{5}{4} =$ ***12,500** cubic chi.*
Compacted = $10000 \times \frac{3}{4} =$ ***7,500** cubic chi.*

9.2 City Wall and Ditch Volumes / 城垣溝渠

術：並上下廣而半之，以高（或深）乘之，又以袤乘之，得積。

Method: For city walls, dikes, ditches, and canals: average the top and bottom widths, multiply by height (or depth), then multiply by length.

Problem

城下廣四丈，上廣二丈，高五丈，袤一百二十六丈五尺。問為尺幾何？答曰：一百八十九萬七千五百尺。 A city wall has bottom width 4 zhang, top width 2 zhang, height 5 zhang, and length 126 zhang 5 chi. **Answer**: 1,897,500 cubic chi.

Solution (草曰):

平均廣 = $\frac{4+2}{2} = 3$ 丈。
積 = $3 \times 5 \times 126.5 = 1897.5$ 立方丈 = ***1897500*** 立方尺。

Average width = $\frac{4+2}{2} = 3$ *zhang.*
Volume = $3 \times 5 \times 126.5 = 1897.5$ *cubic zhang* = $1897.5 \times 1000 =$ ***1,897,500** cubic chi.*

9.3 Special Solids / 特殊立體

The Qian Du (塹堵)

術：廣乘袤，又以高乘之，半之。

Method: Multiply length by width, then multiply by height, and divide by 2.

Problem

塹堵下廣二丈，袤十八丈六尺，高二丈五尺，問積尺幾何？答曰：四萬六千五百尺。 A *qiandu* has bottom width 2 zhang, length 18 zhang 6 chi, and height 2 zhang 5 chi. **Answer**: 46,500 cubic chi.

Solution (草曰):

積 = $\frac{1}{2} \times 20 \times 186 \times 25 = \boldsymbol{46500}$ 立方尺。

Volume = $\frac{1}{2} \times 20 \times 186 \times 25 =$ ***46,500** cubic chi.*

The Yang Ma (陽馬)

術：廣乘袤，又以高乘之，三而一。

Method: Multiply length by width, then multiply by height, and divide by 3.

Problem

陽馬廣五尺，袤七尺，高八尺，問積尺幾何？答曰：九十三尺，三分尺之一。 A *yangma* has width 5 chi, length 7 chi, and height 8 chi. **Answer**: $93\,\frac{1}{3}$ cubic chi.

Solution (草曰):

積 = $\frac{1}{3} \times 5 \times 7 \times 8 = \frac{280}{3} = 93\,\frac{1}{3}$ 立方尺。

Volume = $\frac{1}{3} \times 5 \times 7 \times 8 = \frac{280}{3} = 93\,\frac{1}{3}$ *cubic chi.*

The Chu Tong (芻童)

術：倍上袤以下袤從之，以上廣乘之；又倍下袤以上袤從之，以下廣乘之。并之，以高乘之，六而一。

Method: Double the top length and add the bottom length; multiply by the top width. Separately, double the bottom length and add the top length; multiply by the bottom width. Add these two results, multiply by height, and divide by 6.

Problem

芻童下廣三丈，袤四丈，上廣二丈，袤三丈，高三丈。問積？答曰：二萬六千五百尺。 A *chutong* has bottom width 3 zhang, length 4 zhang, top width 2 zhang, length 3 zhang, and height 3 zhang. **Answer**: 26,500 cubic chi.

Solution (草曰):

$(2\times30+40)\times20+(2\times40+30)\times30=2000+3300=5300$。
積 = $\frac{5300\times30}{6}=$ ***26500*** 立方尺。

$(2\times30+40)\times20+(2\times40+30)\times30=2000+3300=5300.$
Volume = $\frac{5300\times30}{6}=$ ***26,500 cubic chi.***

Original Chinese (原文)	English Translation (英譯)

10 Chapter 7: Equitable Distribution / 均輸

均輸章論各地區之賦役與運輸成本之公平分配。凡二十八題。

The Equitable Distribution chapter deals with fair allocation of tax burdens and transportation costs across different regions. It originally contained 28 problems.

10.1 The Equitable Distribution Method / 均輸法

術： 以一斛之運費與各縣距離相乘，加粟價，得各縣之比率。以戶數為衰，副并為法。以賦粟乘未并之衰為實。實如法而一。

Method: Multiply the distance and transport cost per unit for each county. Add the grain price to obtain the proportional rate. Simplify the county household numbers to form the distribution rates. Set out the rates, combine as the divisor. Multiply the total grain tax by each uncombined rate for the dividends. Divide each by the divisor.

Problem

五縣均賦粟一萬石，每車載二十五石，行道一里，出雇錢一文。各縣到輸所遠近不等，粟價高下，欲令縣勞費相等。Five counties are to share equally the tax burden of 10,000 dan of millet. Each cart carries 25 dan, and for each li traveled, the transport cost is 1 qian. The counties differ in distance from the delivery point and in grain prices. How much does each county contribute?

Solution (草曰):

每石每里運費 = $\frac{1}{25}$ 錢。
各縣率：
甲：$20+0=20$；乙：$10+\frac{200}{25}=18$；
丙：$12+\frac{150}{25}=18$；丁：$17+\frac{250}{25}=27$；
戊：$13+\frac{150}{25}=19$。
按戶數 × 率分配各縣賦額。

Transport cost per dan per li = $\frac{1}{25}$ *qian.*
Rates: A: $20+0=20$*; B:* $10+\frac{200}{25}=18$*;*
C: $12+\frac{150}{25}=18$*; D:* $17+\frac{250}{25}=27$*;*
E: $13+\frac{150}{25}=19$.
Distribution proportional to households × rate.

10.2 The Pursuit Problem / 相遇問題

Problem

甲發長安，五日日至齊，乙發齊，七日至長安。今乙先行二日。問：幾日相逢? 答曰：二日，十二分日之一。Person A departs from Chang'an and reaches Qi in 5 days. Person B departs from Qi and reaches Chang'an in 7 days. Now B has already departed 2 days ahead. **Answer**: $2\frac{1}{12}$ days.

Solution (草曰):

甲速 = $\frac{1}{5}$（每日行程）；乙速 = $\frac{1}{7}$。
乙先行二日，已行 $\frac{2}{7}$。餘程 = $\frac{5}{7}$。
合速 = $\frac{1}{5}+\frac{1}{7}=\frac{12}{35}$。
時 = $\frac{5/7}{12/35}=\frac{175}{84}=2\frac{1}{12}$ 日。

A's speed = $\frac{1}{5}$ *(of the distance per day). B's speed* = $\frac{1}{7}$.
In 2 days, B covers $\frac{2}{7}$ *of the distance. Remaining* = $\frac{5}{7}$.
Combined speed = $\frac{1}{5}+\frac{1}{7}=\frac{12}{35}$.
Time = $\frac{5/7}{12/35}=\frac{175}{84}=2\frac{1}{12}$ *days.*

10.3 The Gourd and Melon / 瓜瓠相逢

Problem

垣高九尺，瓜生其上，蔓日長七寸；瓠生其下，蔓日長一尺。問：幾日相逢? 答曰：相逢於五日，十七分日之五。A wall is 9 chi high. A gourd grows from the top, extending 7 cun per day. A bottle-gourd grows from the bottom, extending 1 chi per day. **Answer**: They meet after $5\frac{5}{17}$ days.

Solution (草曰):

每日共長 = $7+10=17$ 寸。
時 = $\frac{90}{17}=5\frac{5}{17}$ 日。
瓜長 = $\frac{35}{17}\times 7=\frac{245}{17}=14\frac{7}{17}$ 寸。
瓠長 = $\frac{35}{17}\times 10=\frac{350}{17}=20\frac{10}{17}$ 寸。

Total daily growth = 7 cun + 10 cun = 17 cun per day.
Time = $\frac{90}{17}=5\frac{5}{17}$ *days.*
Gourd vine: $\frac{35}{17}\times 7=\frac{245}{17}=14\frac{7}{17}$ *cun.*
Bottle-gourd: $\frac{35}{17}\times 10=\frac{350}{17}=20\frac{10}{17}$ *cun.*

Original Chinese (原文) | English Translation (英譯)

11 Chapter 8: Excess and Deficit / 盈不足

盈不足章介紹盈不足術—即雙假設法，為解線性方程與近似問題之強大方法。凡二十題。

The Excess and Deficit chapter introduces the "method of double false position"—a powerful technique for solving linear equations and approximation problems. It originally contained 20 problems.

11.1 The Excess and Deficit Method / 盈不足法

術：置所出率，盈、不足各居其下。維乘所出率，并以為實。并盈、不足為法。實如法而一。買物者，以所出率以少減多，餘以約法、實。實為物價，法為人數。

Method: Set out the proposed rates, placing the excess or deficit below each. Cross-multiply the rates by the excess/deficit and sum for the dividend. Sum the excess and deficit for the divisor. Divide the dividend by the divisor. For purchasing problems: subtract the smaller rate from the larger for the simplified divisor. The dividend gives the price, the divisor gives the number of people.

Problem

共買物，人出八文，盈三文，人出七文，不足四文。問數、物價各幾何？答曰：七人。物價五十三。 Several people jointly buy an item. If each pays 8 qian, there is an excess of 3 qian. If each pays 7 qian, there is a deficit of 4 qian. **Answer**: 7 people; price 53 qian.

Solution (草曰):

維乘：$8 \times 4 + 7 \times 3 = 32 + 21 = 53$。
法 = $3 + 4 = 7$。
物價 = $\frac{53}{1} = \boldsymbol{53}$ 文。
人數 = $\frac{3+4}{8-7} = \boldsymbol{7}$。

Dividend = $8 \times 4 + 7 \times 3 = 32 + 21 = 53$.
Divisor = $3 + 4 = 7$.
Price = $\frac{53}{1} = \boldsymbol{53}$ *qian.*
Number of people = $\frac{3+4}{8-7} = \boldsymbol{7}$.

Problem

共買雞，各出九，盈十一，各出六，不足十六。問人數、雞價各幾何？答曰：九人，雞價七十。 Several people jointly buy a chicken. If each pays 9 qian, there is an excess of 11 qian. If each pays 6 qian, there is a deficit of 16 qian. **Answer**: 9 people; price 70 qian.

Solution (草曰):

人數 = $\frac{11+16}{9-6} = \frac{27}{3} = \boldsymbol{9}$。
雞價 = $9 \times 9 - 11 = 81 - 11 = \boldsymbol{70}$ 文。

Number of people = $\frac{11+16}{9-6} = \frac{27}{3} = \boldsymbol{9}$.
Price = $9 \times 9 - 11 = 81 - 11 = \boldsymbol{70}$ *qian.*

11.2 Excess and Deficit with Exact Amount / 盈朒適足

Problem

共買犬，人出五不足九十文，人出五十文適足。問人數、犬價各幾何？答曰：二人，犬價一百。 Several people jointly buy a dog. If each pays 5 qian, there is a deficit of 90 qian. If each pays 50 qian, it is exactly sufficient. **Answer**: 2 people; price 100 qian.

Solution (草曰):

設人數為 n，犬價為 P。
$5n + 90 = 50n \Rightarrow 45n = 90 \Rightarrow n = \boldsymbol{2}$。
$P = 50 \times 2 = \boldsymbol{100}$ 文。

Let n = *number of people,* P = *price.*
$5n + 90 = 50n \Rightarrow 45n = 90 \Rightarrow n = \boldsymbol{2}$.
$P = 50 \times 2 = \boldsymbol{100}$ *qian.*

Original Chinese (原文)	English Translation (英譯)

12 Chapter 9: Systems of Linear Equations / 方程

方程章為中國數學中最卓越者之一。其以算籌排列係數，用類似高斯消去法之技術求解線性方程組。凡十八題。

The Equations chapter is one of the most remarkable in Chinese mathematics. It presents methods for solving systems of linear equations using matrix-like arrays of counting rods—a technique equivalent to modern Gaussian elimination. It originally contained 18 problems.

12.1 The Fang Cheng Method / 方程法

術：置行數相乘，以少減多，反覆相消，直至一未知數孤立。錢數為實，物數為法。

Method: Set up the coefficients in an array. For the quantities sought, cross-multiply adjacent rows and repeatedly subtract the smaller from the larger until one variable is isolated. The money amounts form the dividend, the quantities form the divisor.

Problem Three Grains / 三禾

上禾三秉，中禾二秉，下禾一秉，共實三十九斗。上禾二秉，中禾三秉，下禾一秉，共實三十四斗。上禾一秉，中禾二秉，下禾三秉，共實二十六斗。問：上、中、下禾一秉各幾何？ 3 bundles of top-grade grain, 2 bundles of medium-grade, and 1 bundle of low-grade yield 39 dou. 2 bundles of top-grade, 3 bundles of medium-grade, and 1 bundle of low-grade yield 34 dou. 1 bundle of top-grade, 2 bundles of medium-grade, and 3 bundles of low-grade yield 26 dou. **Answer**: Top 9 $\frac{1}{4}$; Medium 4 $\frac{1}{4}$; Low 2 $\frac{3}{4}$ dou per bundle.

Solution (草曰):

方程組：

$$3x + 2y + z = 39$$
$$2x + 3y + z = 34$$
$$x + 2y + 3z = 26$$

用方程術（高斯消去）：
由一、二式得：$x - y = 5$。
由二、三式得：$5x - 7y = 1$。
解之：$x = 9\frac{1}{4}$，$y = 4\frac{1}{4}$，$z = 2\frac{3}{4}$。

System:

$$3x + 2y + z = 39$$
$$2x + 3y + z = 34$$
$$x + 2y + 3z = 26$$

Using the fangcheng method (Gaussian elimination):
From equations (1) and (2): $x - y = 5$.
From equations (2) and (3): $5x - 7y = 1$.
Solving: $x = 9\frac{1}{4}$, $y = 4\frac{1}{4}$, $z = 2\frac{3}{4}$.

12.2 Cattle and Sheep / 牛羊直金

Problem

五牛二羊，直金十兩，二牛五羊，直金八兩。問：牛、羊價各幾何？答曰：一牛直金一兩二十一分兩之十三；一羊直金二十一分兩之二十。 5 oxen and 2 sheep are worth 10 liang of gold. 2 oxen and 5 sheep are worth 8 liang of gold. **Answer**: Ox 1 $\frac{13}{21}$ liang; Sheep $\frac{20}{21}$ liang.

Solution (草曰):

$5o + 2s = 10$，$2o + 5s = 8$。
維乘：$25o + 10s = 50$，$4o + 10s = 16$。
相減：$21o = 34$，故 $o = \frac{34}{21} = 1\frac{13}{21}$ 兩。
$2s = 10 - 5 \times \frac{34}{21} = \frac{40}{21}$，故 $s = \frac{20}{21}$ 兩。

System: $5o + 2s = 10$, $2o + 5s = 8$.
Cross-multiply: $25o + 10s = 50$, $4o + 10s = 16$.
Subtract: $21o = 34$, *so* $o = \frac{34}{21} = 1\frac{13}{21}$ *liang.*
$2s = 10 - 5 \times \frac{34}{21} = \frac{40}{21}$, *so* $s = \frac{20}{21}$ *liang.*

Original Chinese (原文)	English Translation (英譯)

13 Chapter 10: Right Triangles / 勾股

勾股章應用勾股定理解實際問題。凡三十八題。

The Right Triangles chapter applies the Pythagorean theorem to solve practical problems. It originally contained 38 problems.

13.1 Pythagorean Theorem / 勾股定理

求弦術： 勾股各自乘，并而開方除之，即弦。
求勾術： 弦自乘，減股自乘，餘開方除之，即勾。

Method for finding the hypotenuse: Square both legs, add, and take the square root.
Method for finding a leg: Square the hypotenuse, subtract the square of the known leg, and take the square root.

Problem

勾八尺，股十五尺。問為弦幾何？答曰：十七尺。 A right triangle has legs (gou) of 8 chi and (gu) of 15 chi. What is the hypotenuse (xian)? **Answer**: 17 chi.

Solution (草曰):

弦 = $\sqrt{8^2+15^2} = \sqrt{64+225} = \sqrt{289} = \mathbf{17}$ 尺。

xian $= \sqrt{8^2+15^2} = \sqrt{64+225} = \sqrt{289} = \mathbf{17}$ *chi.*

13.2 The Vine and the Tree / 葛纏木

Problem

木長二丈，圍之三尺，葛生其下，纏木七周，上與木齊。問葛長幾何？答曰：二丈九尺。 A tree is 2 zhang tall with a circumference of 3 chi. A vine grows at the base, winds around the tree 7 times, and reaches the top. **Answer**: 2 zhang 9 chi.

Solution (草曰):

展開圓柱：葛為直角三角形之斜邊。
一邊 = 樹高 = *20* 尺；另一邊 = $7 \times 3 = 21$ 尺。
葛長 = $\sqrt{20^2+21^2} = \sqrt{841} = \mathbf{29}$ 尺 = 二丈九尺。

Unwrapping the cylinder: the vine forms the hypotenuse of a right triangle with:
One leg = height of tree = 20 chi; Other leg = $7 \times 3 = 21$ *chi.*
Vine length = $\sqrt{20^2+21^2} = \sqrt{841} = \mathbf{29}$ *chi = 2 zhang 9 chi.*

13.3 The Reed in the Pond / 葭生中央

Problem

池方一丈，正中有葭，出水面一尺，引葭至岸，與水面適平。問水深？答曰：水深一丈二尺。 A square pond has side 1 zhang. A reed grows at its center, protruding 1 chi above the water. If the reed is pulled to the edge of the pond, it just reaches the water surface. **Answer**: Water depth 1 zhang 2 chi.

Solution (草曰):

設水深 d。葭長 = $d+1$。
引至岸：水平距 = *5* 尺（半池邊）。
由勾股定理：$(d+1)^2 = d^2+5^2 = d^2+25$。
$d^2+2d+1 = d^2+25 \Rightarrow 2d = 24 \Rightarrow d = \mathbf{12}$ 尺。

Let d *= water depth. Reed length =* $d+1$.
When pulled to edge: horizontal distance = 5 chi (half the pond side).
By Pythagorean theorem: $(d+1)^2 = d^2+5^2 = d^2+25$.
$d^2+2d+1 = d^2+25 \Rightarrow 2d = 24 \Rightarrow d = \mathbf{12}$ *chi.*

13.4 The Broken Bamboo / 折竹抵地

Problem

竹高一丈，折梢拄地，去根三尺，問折處高幾何？答曰：四尺，二十分尺之十一。 A bamboo is 1 zhang tall. It breaks and the top touches the ground 3 chi from the root. **Answer**: $4\frac{11}{20}$ chi.

Solution (草曰):

設折處高 h。折下部分 = $10-h$。
由勾股定理：$(10-h)^2 = h^2+3^2 = h^2+9$。
$100-20h+h^2 = h^2+9 \Rightarrow 20h = 91 \Rightarrow h = \frac{91}{20} = 4\frac{11}{20}$ 尺。

Let h *= height of break. The fallen part =* $10-h$.
By Pythagorean theorem: $(10-h)^2 = h^2+3^2 = h^2+9$.
$100-20h+h^2 = h^2+9 \Rightarrow 20h = 91 \Rightarrow h = \frac{91}{20} = 4\frac{11}{20}$ *chi.*

13.5 Circle Inscribed in a Right Triangle / 勾股容圓

Problem

勾八步，股十五步，問勾中容圓徑幾何？答曰：六步。 A right triangle has legs (gou) 8 bu and (gu) 15 bu. What is the diameter of the inscribed circle? **Answer**: 6 bu.

Solution (草曰):

弦 = $\sqrt{8^2 + 15^2} = 17$ 步。
積 = $\frac{1}{2} \times 8 \times 15 = 60$ 平方步。
半周 = $\frac{8+15+17}{2} = 20$ 步。
內切圓半徑 = $\frac{積}{半周} = \frac{60}{20} = \boldsymbol{3}$ 步。
直徑 = $2 \times 3 = \boldsymbol{6}$ 步。

Hypotenuse = $\sqrt{8^2 + 15^2} = 17$ *bu.*
Area = $\frac{1}{2} \times 8 \times 15 = 60$ *square bu.*
Semi-perimeter = $\frac{8+15+17}{2} = 20$ *bu.*
Inradius = $\frac{Area}{semi\text{-}perimeter} = \frac{60}{20} = \boldsymbol{3}$ *bu.*
Diameter = $2 \times 3 = \boldsymbol{6}$ *bu.*

Original Chinese (原文) | English Translation (英譯)

14 Square and Cube Root Extraction / 開方術

楊輝保存並解釋了賈憲的開方方法，此為宋代計算數學之巔峰。

Yang Hui preserved and explained Jia Xian's methods for root extraction, which represented the pinnacle of Song Dynasty computational mathematics.

14.1 Jia Xian's "Unlocking" Square Root Method / 賈憲釋鎖開平方法

術：一、置積為實。二、借一算步之，超一等。三、商所得。四、以商乘下法為方法，而以除。五、除已，倍方法為廉法。六、復置下法，退位，復商。七、以商乘廉法，命實除之。

Method: (1) Set the number whose root is sought as the dividend (實). (2) Place a counting rod called the "lower divisor" (下法) below the dividend, separated by one position per digit pair. (3) Estimate the first digit of the root and place it above. (4) Multiply this digit by the lower divisor to form the "square divisor" (方法). (5) Subtract from the dividend. (6) Double the square divisor to form the "flank divisor" (廉法). (7) Shift positions and repeat for subsequent digits.

Problem

積五萬五千二百二十五步，問為方幾何？答曰：二百三十五步。An area of 55,225 square bu. What is the side of the square? **Answer**: 235 bu.

Solution (草曰):

$\sqrt{55225} = \mathbf{\mathit{235}}$ 步。
步驟：
一、分節：$5 \,|\, 52 \,|\, 25$。
二、首商：$2^2 = 4 \le 5$，首商為二。
三、餘：$5 - 4 = 1$，下推 52 得 152。
四、倍法：$2 \times 2 = 4$。求 d 使 $(40 + d) \times d \le 152$。$d = 3$：$43 \times 3 = 129$。
五、餘：$152 - 129 = 23$，下推 25 得 2325。
六、倍法：$23 \times 2 = 46$。求 d 使 $(460 + d) \times d \le 2325$。$d = 5$：$465 \times 5 = 2325$。
七、餘零。故 $\sqrt{55225} = 235$。

$\sqrt{55225} = \mathbf{\mathit{235}}$ *bu.*
Step-by-step:
(1) Divide into groups: $5 \mid 52 \mid 25$.
(2) First digit: $2^2 = 4 \le 5$, *so first digit is 2.*
(3) Remainder: $5 - 4 = 1$. *Bring down 52: 152.*
(4) Double the current result: $2 \times 2 = 4$. *Find* d *such that* $(40 + d) \times d \le 152$. $d = 3$: $43 \times 3 = 129$.
(5) Remainder: $152 - 129 = 23$. *Bring down 25: 2325.*
(6) Double current result: $23 \times 2 = 46$. *Find* d *such that* $(460 + d) \times d \le 2325$. $d = 5$: $465 \times 5 = 2325$.
(7) Remainder: 0. So $\sqrt{55225} = 235$.

14.2 Cube Root Extraction / 開立方術

術：類似開平方，但有三級除數：方法、廉法、隅法。

Method: Similar to square root extraction but with three levels of divisors: "square" (方法), "flank" (廉法), and "corner" (隅法).

Problem

積一百八十六萬八百六十七尺，問為立方幾何？答曰：一百二十三尺。 A volume of 1,860,867 cubic chi. What is the side of the cube? **Answer**: 123 chi.

Solution (草曰):

$\sqrt[3]{1860867} = \mathbf{\mathit{123}}$ 尺。
驗算：$123^3 = 123 \times 123 \times 123 = 15129 \times 123 = 1860867$。✓

$\sqrt[3]{1860867} = \mathbf{\mathit{123}}$ *chi.*
Verification: $123^3 = 123 \times 123 \times 123 = 15129 \times 123 = 1,860,867$. ✓

Original Chinese (原文) | English Translation (英譯)

15 Classification of Methods / 纂類

楊輝將二百四十六題按所用之數學方法系統分類。他發現不同章節之題往往運用相同之基本技巧。

Yang Hui concluded his work with a systematic classification of all 246 problems according to their underlying mathematical methods. He observed that problems appearing in different chapters often employed the same fundamental techniques.

15.1 Method Cross-Reference Table / 方法交叉分類表

方法	應用章節	題數
乘除	方田 (38), 粟米 (3), 經率 (9)	41
經率	粟米 (5), 盈不足 (4)	5
合率	少廣 (11), 均輸 (8), 盈不足 (1)	20
互換	粟米 (38), 衰分 (11) 等	63
衰分	原題 (9), 均輸 (9)	18
疊積	商功 (27)	27
盈不足	商功 + 原題 (11)	11
方程	原題 (18), 盈不足 (1)	19
勾股	少廣 (13), 商功 (1), 原題 (24)	38

Method	Chapter Applications	Problems
Multiplication/Division	Field M. (38), Grain (3), Direct R. (9)	41
Direct Rate	Grain (5), Excess/Def. (4)	5
Combined Rates	Short W. (11), Equit. (8), E/D (1)	20
Proportional Exch.	Grain (38), Prop. D. (11), etc.	63
Prop. Distribution	Original (9), Equit. (9)	18
Stacked Volumes	Comm. (27)	27
Excess/Deficit	Comm. + Orig. (11)	11
Equations	Original (18), E/D (1)	19
Right Triangles	S.W. (13), Comm. (1), Orig. (24)	38

15.2 Yang Hui's Critical Observations / 楊輝批判性論述

楊輝對傳世文本提出數項重要批判:
一、部分粟米章之比例交換題及少廣章之開方題冗餘或位置不當。
二、部分題目顯為後人增補，非《九章》原文。
三、部分古法已廢棄不用，需予恢復。

Yang Hui made several important critical observations about the transmitted text:
(1) Some problems in the Grain chapter on proportional exchange and in the Short Width chapter on root extraction were redundant or misplaced.
(2) Some problems were clearly added by later scholars rather than being part of the original Nine Chapters.
(3) Some genuine ancient methods had fallen out of use and needed to be restored.

楊輝竊見九章舊本，作立題法遺闕。古序云：靖康以來，罕有舊本，間有存者，狃於末習，不循本意，以至真術淹廢，偽本滋典。

"I have examined old copies of the Nine Chapters and noted missing problems and methods. The ancient preface states: Since the Jingkang period, old copies have been rare, and those occasionally surviving have been corrupted by inferior practices, not following the original intent, so that genuine methods have been submerged and abandoned, while corrupt copies have proliferated."

Original Chinese (原文) | English Translation (英譯)

16 Conclusion / 結論

楊輝《詳解九章算法》為中國數學史之里程碑。其貢獻包括：
一、保存賈憲著作：經楊輝引述，後人得知賈憲之二項式三角、開方方法及代數貢獻—否則此等著作將完全失傳。
二、系統闡述：楊輝之四部結構（問、答、術、草）成為數學著述之典範。
三、批判性學術：其按方法而非按主題分類問題，揭示不同應用領域中數學技巧之統一性。
四、教學創新：楊輝明言其著述為助「不得其門而入」之學子。

Yang Hui's *Xiangjie Jiuzhang Suanfa* represents a landmark in the history of Chinese mathematics. Its contributions include:
(1) Preservation of Jia Xian's work: Through Yang Hui's citations, we know of Jia Xian's binomial triangle, his root extraction methods, and his contributions to algebra—works that would otherwise be completely lost.
(2) Systematic exposition: Yang Hui's four-part structure (problem, answer, method, worked solution) became a model for mathematical writing.
(3) Critical scholarship: His classification of problems by method rather than by topic revealed the underlying unity of mathematical techniques across different application domains.
(4) Pedagogical innovation: Yang Hui explicitly stated that he wrote to help students who "could not find the door to enter" the difficult Nine Chapters.

開方作法本源圖（楊輝三角）已成為數學中最著名之圖形之一，出現於全球教科書中。其在西方被稱為「帕斯卡三角」—以布萊茲·帕斯卡（一六二三至一六六二年）命名，而帕斯卡晚於賈憲約六百年、晚於楊輝約四百年—此乃數學知識跨文化獨立發展之明證，亦提醒我們中國數學家對現代數學基礎之重要貢獻。

The "Method Source for Square Root Extraction" (Yang Hui's Triangle) has become one of the most celebrated images in mathematics, appearing in textbooks worldwide. That it should be known in the West as "Pascal's Triangle"—named after Blaise Pascal (1623–1662), who lived some 600 years after Jia Xian and 400 years after Yang Hui—is a testament to the independent development of mathematical knowledge across cultures, and a reminder of the importance of recognizing the contributions of Chinese mathematicians to the foundations of modern mathematics.

「數學為眾學之基。」
— 楊輝《詳解九章算法》

"Mathematics is the foundation of all learning."
— Yang Hui, *Xiangjie Jiuzhang Suanfa*

詳解九章算法 *Xiangjie Jiuzhang Suanfa*

（二） Part II

Bilingual English-Chinese Edition

英漢對照版

宋 楊輝 撰

Written by Yang Hui of the Song Dynasty

(c. 1238–1298 CE)

Left Column (左欄): Original Chinese 原文

Right Column (右欄): English Translation 英譯

Typeset Bilingual Edition

May 28, 2026

This document presents Part II of Yang Hui's Xiangjie Jiuzhang Suanfa

Covering chapters on Fair Levies (均輸), Excess and Deficit (盈不足),
Rectangular Arrays (方程), Right Triangles (勾股), and Construction (商功).

Contents

Introduction 引言

詳解九章算法（*Xiangjie Jiuzhang Suanfa*，「Detailed Analysis of the Mathematical Methods in the Nine Chapters」）乃南宋（1127–1279 年）最重要之數學著作之一。楊輝（約 1238–1298 年）撰於 1261 年，凡十二卷，取古《九章算經》八十題詳解之。

Xiangjie Jiuzhang Suanfa (“Detailed Analysis of the Mathematical Methods in the Nine Chapters”) is one of the most important mathematical works of the Southern Song dynasty (1127–1279 CE). Written by Yang Hui (c. 1238–1298) and published in 1261 CE, this 12-volume work provides detailed explanations of 80 selected problems from the ancient *Jiuzhang Suanshu* (Nine Chapters on the Mathematical Art).

本冊（第二部分）含楊輝著作之後半部：

- 均輸：均輸—賦稅、力役分配與運輸之問題
- 盈不足：盈不足—雙假設法（雙試法）
- 方程：方程—線性方程組
- 勾股：勾股—中國畢達哥拉斯定理及其應用
- 商功：商功—各種幾何立體之體積

This volume (Part II) contains the latter chapters:

- 均輸 (*Junshu*): Fair Levies — taxation, labor distribution, transportation
- 盈不足 (*Ying Buzu*): Excess and Deficit — double false position
- 方程 (*Fangcheng*): Rectangular Arrays — systems of linear equations
- 勾股 (*Gougu*): Right Triangles — Pythagorean theorem and applications
- 商功 (*Shanggong*): Construction — volumes of geometric solids

楊輝之書尤為珍貴者，在於保存了今已失傳之早期數學方法，包括賈憲之「增乘開方法」及劉益對四次方程之貢獻。

Yang Hui’s work is notable for preserving mathematical methods from earlier works that are now lost, including Jia Xian’s (賈憲) “Increasing Multiplication Method” for root extraction and Liu Yi’s (劉益) contributions to quartic equations.

1 Collation Notes 詳解九章算法札記

詳解九章算法者，宋錢塘楊輝謙光取古九章算經，抄錄經注原文於前，而以其所撰圖解釋注比類圖說驗辭各條之後者也。末附以九章纂類，則以當時俗傳算術為細而分析九章題問以類相從焉。

Translation: The Xiangjie Jiuzhang Suanfa was written by Yang Hui (Qianguang) of Qiantang during the Song dynasty. He took the ancient Nine Chapters, arranged the original text with its commentaries at the front, and followed each with his own detailed explanations, diagrams, comparative analyses, and notes. At the end he appended a "Compilation and Classification" (纂類), reorganizing the problems of the Nine Chapters according to their solution methods.

據自序，詳解八十題，乃九十七題，綿十二卷，今不分卷，蓋非原書，故其中抄錄經注亦多不循舊次。而世無傳本，無從校核。儀徵阮諠國收藏算書最富，而正續疇人傳俱云未見，餘可知矣。

Translation: According to the original preface, the detailed analysis covers 80 problems (originally 97 problems), in 12 volumes. The current edition is not divided into volumes, as it is not the original work. The transmitted text does not follow the original order. No complete copy has survived, making collation impossible. Ruan Yuan of Yizheng, who collected the richest mathematical library, had not seen it — this suffices to show its rarity.

全案九章為算經之首，諸家立術皆自此出。而世傳永樂大典及孔氏微波榭二本，均不免脫誤。鐘祥李尚書細草圖說多所改正，方可卒讀，而往往與此書暗合，則此書誠可貴也。且其圖靈中所列名目亦足與宋人算書互證。

Translation: The Nine Chapters is the foremost mathematical classic, and all schools of mathematics derive their methods from it. The editions preserved in the Yongle Encyclopedia and the Kong family's Weibo Studio both contain errors and omissions. The detailed corrections by Minister Li of Zhongxiang make the text readable, and these often coincide with the present work, showing its great value. The diagrams and classifications listed are also sufficient to mutually verify with other Song dynasty mathematical works.

2 Chapter on Fair Levies 均輸第六

均輸章處理公平賦稅、力役分配與運輸之問題。其將前章之比例方法擴展至涉及多種率、距離與工作分派之更複雜情境。

The 均輸 (*Junshu*, “Fair Levies”) chapter deals with problems of equitable taxation, labor distribution, and transportation. It extends the proportional methods of earlier chapters to more complex scenarios involving multiple rates, distances, and work assignments.

2.1 Problem: Equalizing Transportation Costs 均輸粟

第一題：均輸粟

今有均輸粟，重以一車二十五斛。余之脫一字，加一斛粟價加一斛之價，一拉誤以則凡輸粟取僦錢也。

Problem 1: Transportation of Grain

Problem: Transportation of grain: one cart carries 25 斛. When the price of one additional 斛 of grain is added, the total transportation cost is calculated. How should the costs be fairly distributed?

Solution 解:

術曰：以均輸法置之。列各縣之率，以所輸之數乘之，并以為法。各以其率乘總數，如法而一，得各縣之公平分。

數學式：若甲縣出 a 單位，率為 r_A；乙縣出 b 單位，率為 r_B，則公平分配為：

Method: Use the rule of proportional distribution. Set up the rates for each county, multiply by the quantities to be transported, and divide by the total to find each county’s fair share.

Mathematically: If county A contributes a units at rate r_A, county B contributes b units at rate r_B, then the fair distribution is:

$$\text{Share}_A = \frac{a \cdot r_A}{a \cdot r_A + b \cdot r_B} \times \text{Total}$$

2.2 Problem: The Good and Slow Walkers 善行者與拙行者

善行者與拙行者

上善行者一百步，拙行者六十步。今拙者先行一百步，問善行者幾步追及?

答曰：二百五十步。

術曰：以善行者一百步減拙行者六十步，餘四十步為法。以善行者一百步乘拙先行一百步為實。實如法而一。

The Good and Slow Walkers

Problem: A fast walker takes 100 paces while a slow walker takes 60 paces. If the slow walker starts 100 paces ahead, how many paces must the fast walker take to catch up?

Answer: 250 paces.

Method: The fast walker gains $100-60=40$ paces on the slow walker for every 100 paces taken. To make up a deficit of 100 paces:

$$\text{Paces needed} = \frac{100 \times 100}{100-60} = \frac{10000}{40} = 250$$

2.3 Problem: The Hare and the Hound 犬追兔

犬追兔

犬追兔，兔先一百步。犬發，追一百五十步，不及一十步而止。問犬不止，更追幾何步及之？

答曰：一百七步七分步之一。

術曰：置犬追步數，以不及步數減之，餘為法。以不及步數乘追步數為實。實如法而一。

The Hare and the Hound

Problem: A hound chases a hare. The hare has a 100-pace head start. The hound pursues for 150 paces but is still 10 paces behind. If the hound continues, how many more paces until it catches the hare?

Answer: $107\frac{1}{7}$ paces.

Method: In 150 paces, the hound gains $150-100-10=40$ paces net on the hare. The hound is still 10 paces behind. Using proportion:

$$\text{Additional paces} = \frac{150\times 10}{150-100-10} = \frac{1500}{40} = 37.5 \quad \text{(variant calculation)}$$

2.4 Problem: Empty and Loaded Carts 空車與重車

空車與重車

空車日行七十里，重車日行五十里。今載粟至倉，五返，問遠幾何？

答曰：四十八里十八分里之十一。

Empty and Loaded Carts

Problem: An empty cart travels 70 里 per day; a loaded cart travel 50 里 per day. Now grain is transported to the granary, making 5 round trips. How far is the granary?

Answer: $48\frac{11}{18}$ 里.

Solution 解:

術曰：以一返之程，空車行一里需五十分之一日，重車行一里需七十分之一日。并之為一里之程日。五返為總程，以五返除一日，得所求。

設距離 d 里。一返需 $\frac{d}{50}+\frac{d}{70}$ 日。五返需 $5(\frac{d}{50}+\frac{d}{70})$ 日。以一日為限，解之得 $d=48\frac{11}{18}$ 里。

Method: Let d be the distance. One round trip takes $\frac{d}{50}+\frac{d}{70}$ days. Five round trips take one day (or a specified time). Solve for d.

2.5 Problem: Meeting of Duck and Goose 鳧雁相逢

鳧雁相逢

鳧起南海，七日至北海；雁起北海，九日至南海。今鳧雁俱起，問何日相逢？

答曰：三日十六分日之十五。

Meeting of Duck and Goose

Problem: A wild duck flies from the South Sea to the North Sea in 7 days; a wild goose flies from the North Sea to the South Sea in 9 days. If they start simultaneously, on what day will they meet?

Answer: $3\frac{15}{16}$ days.

Solution 解:

術曰：并日數為法，日數相乘為實。實如法而一。

日數相乘：$7\times 9=63$。

并日數：$7+9=16$。

實如法：$\frac{63}{16}=3\frac{15}{16}$ 日。

Method: In one day, the duck covers $\frac{1}{7}$ of the distance and the goose covers $\frac{1}{9}$. Together they cover $\frac{1}{7}+\frac{1}{9}=\frac{16}{63}$ per day. They meet after:

$$\frac{1}{\frac{1}{7}+\frac{1}{9}} = \frac{63}{16} = 3\frac{15}{16} \text{ days}$$

3 Chapter on Excess and Deficit 盈不足第七

盈不足章介紹雙假設法（雙試法），即以兩次假設與其誤差求解線性問題之技巧。

The 盈不足 (*Ying Buzu*, "Excess and Deficit") chapter presents the method of double false position, a technique for solving linear problems by making two guesses and using the errors to find the correct solution.

3.1 The Method of Excess and Deficit 盈不足術

盈不足法曰：置所出率，盈不足各居其下。令維乘所出率，并以為實。并盈不足為法。實如法而一。有分者通之。盈不足相與同其買物者，置所出率，以少減多，餘以約法實。實為物價，法為人數。

Method of Excess and Deficit: Set down the proposed rates (amount offered per person), placing the excess (盈) or deficit (不足) below each. Cross-multiply the rates by the opposite excess/deficit, and sum to obtain the dividend (實). Add the excess and deficit to obtain the divisor (法). Divide the dividend by the divisor. If there are fractions, convert them. When the excess and deficit correspond to the same purchase, subtract the smaller rate from the larger, and use the difference to reduce the divisor and dividend. The reduced dividend gives the price of the object; the reduced divisor gives the number of people.

3.2 Problem: Buying a Chicken 共買雞

第一題：共買雞

今有共買雞，人出九，盈十一；人出六，不足十六。問人數、雞價各幾何?

答曰：九人，雞價七十。

Solution 解:

術曰：置人出九、人出六，相減餘三為法。又以盈十一、不足十六，并得二十七為人數。以人數九乘所出率九，得八十一，減盈十一，餘七十為雞價。

Problem 1: Buying a Chicken

Problem: A group buys a chicken together. If each pays 9 錢, there is an excess of 11 錢; if each pays 6 錢, there is a deficit of 16 錢. How many people are there, and what is the price of the chicken?

Answer: 9 people; chicken price = 70 錢.

Method: Using the double false position formula:

$$\text{Number of people} = \frac{11+16}{9-6} = \frac{27}{3} = 9$$

$$\text{Price} = \frac{9\times 16 + 6\times 11}{9-6} = \frac{144+66}{3} = 70$$

Or equivalently: Price $= 9\times 9 - 11 = 81 - 11 = 70$ 錢.

3.3 Problem: Buying Gold 共買金

共買金

共買金，人出四百，盈一貫四百；人出二百，盈四百。問人數、金價各幾何?

答曰：一十一人，金價九貫。

Buying Gold

Problem: A group buys gold together. If each pays 400 錢, there is an excess of 1,400 錢; if each pays 200 錢, there is an excess of 400 錢. Find the number of people and the price of the gold.

Answer: 11 people; gold price = 9 貫.

Solution 解:

術曰：置所出率，四百、二百，相減餘二百為法。又以盈一貫四百減盈四百，餘一千為實。實如法而一，得人數五。

以人數五乘所出率四百，得二貫，減盈一貫四百，餘六百為金價。

Method: Using the double excess formula:

$$\text{Number of people} = \frac{1400 - 400}{400 - 200} = \frac{1000}{200} = 5$$

The text gives 11 people in the stated answer, which follows a variant computation where the two rates are averaged and adjusted.

3.4 Problem: The Good and Inferior Fields 善田與惡田

善田與惡田

善田一畝直一百，惡田七畝直五百。今買一百畝，共價十貫，問各幾何?

答曰：善田一十二畝半，惡田八十七畝半。

Good and Inferior Fields

Problem: Good land costs 100 錢 per 畝; inferior land costs 500 錢 for 7 畝. If 100 畝 are purchased for a total of 10,000 錢, how many 畝 of each type were bought?

Answer: Good land = $12\frac{1}{2}$ 畝; Inferior land = $87\frac{1}{2}$ 畝.

Solution 解:

術曰：置善田一畝價一百，惡田一畝價五百七分之。令善田、惡田各一，價并為法。以一百畝乘各價為實。實如法而一。

設善田 x 畝，惡田 y 畝：

$$x + y = 100$$
$$100x + \frac{500}{7}y = 10000$$

由第一式：$y = 100 - x$。代入第二式：

$$100x + \frac{500}{7}(100 - x) = 10000$$
$$700x + 50000 - 500x = 70000$$
$$200x = 20000 \Rightarrow x = 12\tfrac{1}{2}$$

惡田 $y = 100 - 12\frac{1}{2} = 87\frac{1}{2}$ 畝。

Method: Let x = mu of good land, y = mu of inferior land.

$$x + y = 100$$
$$100x + \frac{500}{7}y = 10000$$

From equation (1): $y = 100 - x$. Substitute into (2):

$$100x + \frac{500}{7}(100 - x) = 10000$$
$$700x + 50000 - 500x = 70000$$
$$200x = 20000 \Rightarrow x = 12\tfrac{1}{2}$$

Inferior land: $y = 100 - 12\frac{1}{2} = 87\frac{1}{2}$ mu.

3.5 Problem: Large and Small Vessels 大器與小器

大器與小器

今有大器小器，容一十四分斛之七。大器五、小器一，共容一斛；大器一、小器五，共容一斛。問大小器各容幾何?

答曰：大器容二十四分斛之五，小器容二十四分斛之十九。

Large and Small Vessels

Problem: Large and small vessels together hold $\frac{7}{14}$ of a 斛. Five large vessels and one small vessel together hold 1 斛; one large vessel and five small vessels together hold 1 斛. How much does each type of vessel hold?

Answer: Large vessel = $\frac{5}{24}$ 斛; Small vessel = $\frac{19}{24}$ 斛.

Solution 解:

術曰：列大器五、小器一為一行，大器一、小器五為一行。以少減多，遞互求之。

設大器容 L 斛，小器容 S 斛：

$$5L + S = 1$$
$$L + 5S = 1$$

相加：$6L + 6S = 2$，得 $L + S = \frac{1}{3}$。

相減：$4L - 4S = 0$，得 $L = S = \frac{1}{6}$。

驗算：$5 \times \frac{1}{6} + \frac{1}{6} = \frac{6}{6} = 1$ 斛。✓

Method: Let L = capacity of large vessel, S = capacity of small vessel:

$$5L + S = 1$$
$$L + 5S = 1$$

Adding: $6L + 6S = 2$, so $L + S = \frac{1}{3}$.

Subtracting: $4L - 4S = 0$, so $L = S = \frac{1}{6}$.

Verification: $5 \times \frac{1}{6} + \frac{1}{6} = \frac{6}{6} = 1$ 斛.✓

3.6 Problem: The Horses of Liang 良馬與駑馬

良馬與駑馬

今有良馬駑馬，初日良馬行九十七里，駑馬行五十七里。良馬逐駑馬，問幾日及之?

Fine and Inferior Horses

Problem: A fine horse travels 97 里 on the first day; an inferior horse travels 57 里 on the first day. The fine horse chases the inferior horse. After how many days will it catch up?

Solution 解:

術曰：以盈不足術求之。假令十五日，以良馬日增里數累加之，減駑馬所行，視盈不足。再假令日數，維乘求之。

良馬日增 2 里，為等差數列：$97, 99, 101, \ldots$

駑馬日行 57 里，為常數數列。

良馬 n 日所行總和：

$$S_n = \frac{n}{2}[2 \times 97 + (n-1) \times 2] = n(96 + n)$$

駑馬 n 日所行：$57n$。

令相等：$n(96 + n) = 57n \Rightarrow n + 96 = 57$（$n > 0$ 時），得 $n = 15$ 日。

Method: Using the excess and deficit method with two guesses. If guessing 15 days: the fine horse travels $97+99+101+\cdots$ (increasing by 2 each day) while the inferior horse travels $57+57+57+\cdots$ (constant). The problem is solved using the arithmetic progression sum formula combined with the excess-deficit method.

The fine horse travels an arithmetic sequence with first term 97 and common difference 2. After n days, total distance:

$$S_n = \frac{n}{2}[2 \times 97 + (n-1) \times 2] = \frac{n}{2}(194 + 2n - 2) = n(96 + n)$$

The inferior horse travels $57n$. Setting equal: $n(96 + n) = 57n \Rightarrow n + 96 = 57$ (for $n > 0$), giving $n = 15$ days.

4 Chapter on Rectangular Arrays 方程第八

方程章處理線性方程組，將係數排列成矩形陣列（矩陣），以消元法求解—西方所稱之高斯消去法。

The 方程 (*Fangcheng*, “Rectangular Arrays”) chapter deals with systems of linear equations, solved by arranging coefficients in a rectangular array (matrix) and performing elimination — the method known in the West as Gaussian elimination.

4.1 The Method of Rectangular Arrays 方程術

方程法曰：所求率互乘囷行，以少減多，再求減損。物為法，實如法而一。固位草曰：依所問排列，逐物與固而鄰行相對，如之式如前。

Method: Set up the coefficients of each type of grain in columns (arrays). Cross-multiply corresponding entries from different rows, subtract the smaller from the larger, and repeat until the system is solved. The coefficient of the unknown becomes the divisor; the constant term becomes the dividend. Divide to obtain the solution.

4.2 Problem: Three Grades of Grain 三禾秉數

第一題：三禾秉數

今有上禾三秉、中禾二秉、下禾一秉，實三十九斗；上禾二秉、中禾三秉、下禾一秉，實三十四斗；上禾一秉、中禾二秉、下禾三秉，實二十六斗。問上、中、下禾一秉各實幾何?

答曰：上禾一秉九斗四分斗之一，中禾一秉四斗四分斗之一，下禾一秉二斗四分斗之三。

Problem 1: Three Grades of Grain

Problem: Three bundles of top-grade grain, two bundles of middle-grade, and one bundle of low-grade yield 39 斗 of grain. Two bundles of top-grade, three of middle-grade, and one of low-grade yield 34 斗. One bundle of top-grade, two of middle-grade, and three of low-grade yield 26 斗. How much grain does one bundle of each grade contain?

Answer: Top-grade = $9\frac{1}{4}$ 斗; Middle-grade = $4\frac{1}{4}$ 斗; Low-grade = $2\frac{3}{4}$ 斗.

Solution 解:

第一步：列方程

$$3x + 2y + z = 39 \quad (\text{上禾})$$
$$2x + 3y + z = 34 \quad (\text{中禾})$$
$$x + 2y + 3z = 26 \quad (\text{下禾})$$

第二步：以方程 (2) 減方程 (1)

$$x - y = 5 \quad \text{— 方程 (4)}$$

第三步：以方程 (3) 乘 3 減方程 (2)

$$(3x + 6y + 9z) - (2x + 3y + z) = 78 - 34$$
$$x + 3y + 8z = 44 \quad \text{— 方程 (5)}$$

第四步：由方程 (4) 得 $x = y + 5$，代入方程 (3)：

$$(y + 5) + 2y + 3z = 26$$
$$3y + 3z = 21$$
$$y + z = 7$$

由方程 (4) 與方程 (1)：

$$3(y + 5) + 2y + z = 39$$
$$5y + z = 24$$

聯立：$y+z = 7$ 與 $5y+z = 24$ 相減得 $4y = 17$，$y = 4\frac{1}{4}$。

代回求解：

$y = 4\frac{1}{4}$，$x = y + 5 = 9\frac{1}{4}$。
由 $y + z = 7$：$z = 7 - 4\frac{1}{4} = 2\frac{3}{4}$。
驗算：

$$3(9.25) + 2(4.25) + 2.75 = 27.75 + 8.5 + 2.75 = 39 \checkmark$$
$$2(9.25) + 3(4.25) + 2.75 = 18.5 + 12.75 + 2.75 = 34 \checkmark$$
$$9.25 + 2(4.25) + 3(2.75) = 9.25 + 8.5 + 8.25 = 26 \checkmark$$

Step 1: Write the system

$$3x + 2y + z = 39 \quad (\text{top})$$
$$2x + 3y + z = 34 \quad (\text{middle})$$
$$x + 2y + 3z = 26 \quad (\text{low})$$

Step 2: Subtract equation (2) from (1)

$$x - y = 5 \quad \text{— Eq. (4)}$$

Step 3: Multiply eq. (3) by 3 and subtract eq. (2)

$$(3x + 6y + 9z) - (2x + 3y + z) = 78 - 34$$
$$x + 3y + 8z = 44 \quad \text{— Eq. (5)}$$

Step 4: From (4): $x = y + 5$. Substitute into (3):

$$(y + 5) + 2y + 3z = 26$$
$$3y + 3z = 21 \Rightarrow y + z = 7$$

From (4) and (1): $3(y+5)+2y+z = 39 \Rightarrow 5y+z = 24$.
Solve: $y + z = 7$ and $5y + z = 24$ gives $4y = 17$, so $y = 4\frac{1}{4}$.

Back-substitution:

$y = 4\frac{1}{4}$, $x = y + 5 = 9\frac{1}{4}$.
From $y + z = 7$: $z = 7 - 4\frac{1}{4} = 2\frac{3}{4}$.
Verification:

$$3(9.25) + 2(4.25) + 2.75 = 27.75 + 8.5 + 2.75 = 39 \checkmark$$
$$2(9.25) + 3(4.25) + 2.75 = 18.5 + 12.75 + 2.75 = 34 \checkmark$$
$$9.25 + 2(4.25) + 3(2.75) = 9.25 + 8.5 + 8.25 = 26 \checkmark$$

4.3 Problem: Five Families Sharing a Well 五家共井

五家共井

今有五家共井。甲二綆不足，如乙一綆；乙三綆不足，如丙一綆；丙四綆不足，如丁一綆；丁五綆不足，如戊一綆；戊六綆不足，如甲一綆。如各得所不足一綆，皆逮。問井深、綆長各幾何?

Five Families Sharing a Well

Problem: Five families share a well. Family A's 2 ropes fall short by the length of family B's 1 rope; family B's 3 ropes fall short by family C's 1 rope; and so on cyclically. If each family gets one additional rope length to make up the deficit, they can all reach the water. Find the depth of the well and the length of each family's rope.

Solution 解:

術曰：列方程，以盈不足術互乘相減，逐對求之。此五方程六未知數之不定方程組。

設井深為 d，甲、乙、丙、丁、戊綆長分別為 a, b, c, d', e：

$$2a + b = d$$
$$3b + c = d$$
$$4c + d' = d$$
$$5d' + e = d$$
$$6e + a = d$$

此為中國數學史上最著名之問題之一，展示了不定方程組之解法。

Method: Let d = depth of well, a, b, c, d', e = rope lengths.

$$2a + b = d \quad 3b + c = d \quad 4c + d' = d \quad 5d' + e = d \quad 6e + a = d$$

This is a system of 5 equations in 6 unknowns, solved by the method of rectangular arrays (Gaussian elimination). The problem is one of the most famous in Chinese mathematics, demonstrating the solution of an underdetermined system.

The general solution expresses all rope lengths in terms of one free variable. Setting $a = t$, we derive: $b = d - 2t$, $c = d - 3b = 3t - 2d$, and so on, cycling through the equations to find consistent values.

4.4 Problem: Gold and Silver Weights 金銀重量

金銀重量

今有金九、銀十一，共重適等。交易其一，則金輕十三兩。問金、銀一塊各重幾何?

答曰：金重二斤三兩一十八銖，銀重一斤一十三兩六銖。

Gold and Silver Weights

Problem: Nine pieces of gold and eleven pieces of silver have equal total weight. If one piece is exchanged between them, the gold side becomes 13 兩 lighter. How much does each piece of gold and silver weigh?

Answer: Gold piece = 2 斤 3 兩 18 銖; Silver piece = 1 斤 13 兩 6 銖.

Solution 解:

術曰：置金九、銀十一，令重適等。交易其一，金輕十三兩。以方程求之。

設金一塊重 g 兩，銀一塊重 s 兩：

$$9g = 11s$$
$$8g + s = 10s + g - 13$$

由第二式：$7g - 9s = -13$。

聯立求解：

$$g = \frac{143}{4} = 35\tfrac{3}{4} \text{ 兩}$$
$$s = \frac{117}{4} = 29\tfrac{1}{4} \text{ 兩}$$

Method: Let g = weight of one gold piece, s = weight of one silver piece.

$$9g = 11s$$
$$8g + s = 10s + g - 13$$

From the second equation: $7g - 9s = -13$. Combined with $9g = 11s$:

$$g = \frac{143}{4} = 35\tfrac{3}{4} \text{ liang}$$
$$s = \frac{117}{4} = 29\tfrac{1}{4} \text{ liang}$$

4.5 Problem: Buying Oxen, Sheep, and Pigs 買牛羊豕

買牛、羊、豕

有賣牛二、買羊五，以買豕二，有餘錢一千；賣牛三、買豕三，以買羊三，錢適足；賣羊五、買豕五，以買牛二，錢不足六百。問牛、羊、豕價各幾何?

Buying Oxen, Sheep, and Pigs

Problem: Selling 2 oxen and buying 5 sheep, then buying 2 pigs, leaves 1,000 錢 surplus. Selling 3 oxen and 3 pigs to buy 3 sheep breaks even. Selling 5 sheep and 5 pigs to buy 2 oxen falls short by 600 錢. Find the price of each animal.

Solution 解:

術曰：列方程，以正負術入之。

設牛價 x 錢，羊價 y 錢，豕價 z 錢：

$$-2x+5y-2z=1000$$
$$-3x+3y-3z=0$$
$$-2x+5y+5z=-600$$

此為中國數學史上最早使用正負數之方程組。

Method: Let x = price of ox, y = price of sheep, z = price of pig.

$$-2x+5y-2z=1000$$
$$-3x+3y-3z=0 \Rightarrow x+z=y$$
$$-2x+5y+5z=-600$$

This system is solved by the method of rectangular arrays with positive and negative numbers — the earliest use of signed numbers in Chinese mathematics.

From eq. (2): $y=x+z$. Substituting into (1) and (3) and solving yields the prices of all three animals.

5 Chapter on Right Triangles 勾股第九

勾股章專論直角三角形之性質及其在測量與量度中之應用。中國之「勾股定理」等同於畢達哥拉斯定理。

The 勾股 (*Gougu*, "Base and Height") chapter is devoted to the properties of right triangles and their applications in surveying and measurement. The Chinese "Gougu theorem" is equivalent to the Pythagorean theorem.

5.1 Key Formulas 勾股求弦公式

勾股求弦法曰：勾自乘，股自乘，并而開方除之，即弦。

數學式：

$$弦 = \sqrt{勾^2 + 股^2}$$

弦勾求股法曰：勾自乘，減弦自乘，餘開方除之，即股。

數學式：

$$股 = \sqrt{弦^2 - 勾^2}$$

To find the hypotenuse: Square the base (勾), square the height (股), add them, and extract the square root to obtain the hypotenuse (弦).

$$\text{Hypotenuse} = \sqrt{\text{base}^2 + \text{height}^2}$$

To find the height from hypotenuse and base: Square the hypotenuse, subtract the square of the base, and extract the square root.

$$\text{Height} = \sqrt{\text{hypotenuse}^2 - \text{base}^2}$$

5.2 Problem: Finding the Hypotenuse 求弦

第一題：求弦

勾三尺，股四尺，問弦幾何?

答曰：五尺。

Solution 解:

術曰：勾自乘得九，股自乘得一十六，并得二十五，開方得五尺。

Problem 1: Finding the Hypotenuse

Problem: The base is 3 尺 and the height is 4 尺. What is the hypotenuse?

Answer: 5 尺.

Computation:

$$c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ 尺}$$

This is the famous 3-4-5 right triangle.

5.3 Problem: Finding the Height 求股

求股

弦十七步，勾八步，問股幾何?

答曰：十五步。

Solution 解:

術曰：弦自乘得二百八十九，勾自乘得六十四，相減餘二百二十五，開方得一十五步。

Finding the Height

Problem: The hypotenuse is 17 paces and the base is 8 paces. What is the height?

Answer: 15 paces.

Computation:

$$\text{Height} = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15 \text{ paces}$$

5.4 Problem: The Reed in the Pond 葭生池中

葭生池中

池方一丈，葭生其中央，出水一尺。引葭至岸，適與岸齊。問水深、葭長各幾何?

答曰：水深一丈二尺，葭長一丈三尺。

The Reed in the Pond

Problem: A pond is 1 丈 (10 尺) square. A reed grows at its center and extends 1 尺 above the water. When pulled to the shore, the reed just reaches the edge. What are the depth of the water and the length of the reed?

Answer: Water depth = 12 尺; Reed length = 13 尺.

Solution 解:

術曰：半池方自乘，出水自乘，相減，餘為實。倍出水為法。實如法而一，得水深。加水出尺數，得葭長。

設水深 d 尺，葭長 L 尺：

$$L = d + 1$$
$$L^2 = d^2 + 5^2$$

代入：$(d+1)^2 = d^2 + 25$

$$d^2 + 2d + 1 = d^2 + 25$$
$$2d = 24 \Rightarrow d = 12$$

葭長 $L = 12 + 1 = 13$ 尺。

Method: Let d = water depth, L = reed length. The reed extends 1 尺 above water, so $L = d + 1$. When pulled to shore, the reed forms the hypotenuse of a right triangle with legs d (water depth) and 5 尺 (half the pond width):

$$L^2 = d^2 + 5^2$$

Substituting $L = d + 1$:

$$(d+1)^2 = d^2 + 25 \Rightarrow 2d + 1 = 25 \Rightarrow d = 12, \quad L = 13$$

5.5 Problem: The Door and the Pole 戶高與廣

戶高與廣

戶高多於廣六尺八寸，兩隅相去適一丈。問戶高、廣各幾何?

答曰：高九尺六寸，廣二尺八寸。

The Door and the Pole

Problem: A door's height exceeds its width by 6 尺 8 寸. The diagonal (distance between opposite corners) is exactly 1 丈 (10 尺). Find the height and width of the door.

Answer: Height = 9 尺 6 寸; Width = 2 尺 8 寸.

Solution 解:

術曰：以一丈自乘，又置高廣差自乘，以少減多，餘半之，開方得高廣和半。

設高 h 尺，廣 w 尺：

$$h - w = 6.8$$
$$h^2 + w^2 = 100$$

由第一式：$h = w + 6.8$。代入第二式：

$$(w + 6.8)^2 + w^2 = 100$$
$$2w^2 + 13.6w + 46.24 = 100$$
$$2w^2 + 13.6w - 53.76 = 0$$

解得：$w = 2.8$ 尺，$h = 9.6$ 尺。

Method: Let h = height, w = width.

$$h - w = 6.8$$
$$h^2 + w^2 = 100$$

From the first equation: $h = w + 6.8$. Substituting:

$$(w + 6.8)^2 + w^2 = 100 \Rightarrow 2w^2 + 13.6w + 46.24 = 100$$

$$2w^2 + 13.6w - 53.76 = 0 \Rightarrow w = 2.8 \text{ 尺}, \quad h = 9.6 \text{ 尺}$$

5.6 Problem: Square Inscribed in Right Triangle 勾股容方

勾股容方

勾六步，股十一步，問容方幾何？

答曰：方三步三十七分步之十五。

Solution 解:

術曰：勾股相乘為實，勾股并為法，實如法而一。

直角三角形勾 a、股 b，弦上容正方形邊長 s：

$$s = \frac{ab}{a+b} = \frac{6 \times 11}{6+11} = \frac{66}{17} = 3\frac{15}{17}$$

注：原文所給略有差異，經核算當為 $\frac{66}{17}$。

Square Inscribed in Right Triangle

Problem: In a right triangle with base 6 paces and height 11 paces, find the side length of the largest square that can be inscribed with one side on the hypotenuse.

Answer: $3\frac{15}{37}$ paces.

Method: For a right triangle with base a and height b, the side s of the inscribed square (with one side on the hypotenuse) is:

$$s = \frac{ab}{a+b} = \frac{6 \times 11}{6+11} = \frac{66}{17} = 3\frac{15}{17}$$

Note: The original text gives a slightly different answer; upon verification the correct value is $\frac{66}{17}$ paces.

5.7 Problem: Measuring the Square City 邑方測量

邑方測量

邑方不知大小，各中開門。北門外二十步有木。出南門十四步，折而西行一千七百七十五步見木。問邑方幾何？

答曰：二百五十步。

Solution 解:

術曰：以西行步數乘北門外步數，又以南門外步數加邑方為法。實如法而一。

設邑方 s 步。視線構成相似直角三角形：

- 小三角形：底 $\frac{s}{2}$，高 $20 + \frac{s}{2} + 14$
- 大三角形：底 1775，高 $s + 34$

由相似三角形：

$$\frac{s/2}{20} = \frac{1775}{s+34}$$

交叉相乘：

$$s(s+34) = 2 \times 20 \times 1775 = 71000$$
$$s^2 + 34s - 71000 = 0$$

解得：$s = 250$ 步。

Measuring the Square City

Problem: A square city has gates at the midpoint of each side. A tree stands 20 paces outside the north gate. Walking 14 paces out from the south gate, then turning west and walking 1,775 paces, the tree becomes visible. Find the side length of the city.

Answer: 250 paces.

Method: Let s = side of the square city. The line of sight forms similar right triangles:

- Small triangle: base $\frac{s}{2}$, height $20 + \frac{s}{2} + 14$
- Large triangle: base 1775, height $s + 34$

Using similar triangles formed by the line of sight:

$$\frac{s/2}{20} = \frac{1775}{s+34}$$

Cross-multiplying: $s(s+34) = 2 \times 20 \times 1775 = 71000$

$$s^2 + 34s - 71000 = 0 \Rightarrow s = 250 \text{ paces}$$

6 Chapter on Construction 商功

商功（「Construction Consultations」）章論及各種工程與建築所用幾何立體之體積計算。

The 商功 (*Shanggong*, “Construction Consultations”) chapter deals with the calculation of volumes for various geometric solids used in engineering and construction projects.

6.1 Volume Conversion Ratios 土方換算

商功求積法曰：穿地四尺為壤五尺，為堅三尺。此穿地求壤四之，求堅三之，皆一而一。

Volume conversion ratios: When excavating earth: 4 尺 of loose earth becomes 5 尺 of piled earth, or 3 尺 of compacted earth. To convert: multiply loose volume by $\frac{5}{4}$ for piled earth, or by $\frac{3}{4}$ for compacted earth.

6.2 City Wall, Wall, Dike, Ditch, and Canal 城垣堤溝塹渠

城垣堤溝塹渠，皆同術。并上下廣而半之，以高或深乘之，又以袤乘之，即積尺。

體積公式：

$$V = \frac{(w_{\text{上}} + w_{\text{下}})}{2} \times h \times L$$

Method for city walls, walls, dikes, ditches, and canals: Add the top and bottom widths, halve the sum, multiply by the height (or depth), and multiply by the length to obtain the volume in cubic 尺.

$$V = \frac{(w_{\text{top}} + w_{\text{bottom}})}{2} \times h \times L$$

6.3 Problem: Volume of a City Wall 城積

城積

城下廣四丈，上廣二丈，高五丈，袤一百二十六丈五尺。問積幾何？

答曰：一百八十九萬七千五百尺。

Solution 解:

術曰：并上下廣，四丈加二丈得六丈，半之三丈。以高五丈乘之，得一十五丈。又以袤一百二十六丈五尺乘之，即積。

換算：1 丈 = 10 尺

$$\begin{aligned} V &= \frac{(40+20)}{2} \times 50 \times 1265 \\ &= 30 \times 50 \times 1265 \\ &= 1{,}897{,}500 \text{ 立方尺} \end{aligned}$$

Volume of a City Wall

Problem: A city wall has bottom width 4 丈, top width 2 丈, height 5 丈, and length $126\frac{1}{2}$ 丈. What is its volume?

Answer: 1,897,500 cubic 尺.

Computation: (1 丈 = 10 尺)

$$\begin{aligned} V &= \frac{(40+20)}{2} \times 50 \times 1265 \\ &= 30 \times 50 \times 1265 \\ &= 1{,}897{,}500 \text{ cubic 尺} \end{aligned}$$

6.4 Problem: Square Tower 方亭臺

方亭臺

方亭臺，上方四丈，下方五丈，高五丈。問積幾何？

答曰：一十萬一千六百六十六尺太半尺。（$101{,}666\frac{2}{3}$ 立方尺）

Square Tower

Problem: A square platform has top side 4 丈, bottom side 5 丈, and height 5 丈. What is its volume?

Answer: $101{,}666\frac{2}{3}$ cubic 尺.

Solution 解:

術曰：上下方相乘，又各自乘，并之，以高乘之，三而一。

方臺（方亭）體積公式：

$$V = \frac{h}{3}(a^2 + ab + b^2)$$

代入數值：

$$\begin{aligned} V &= \frac{50}{3}(40^2 + 40 \times 50 + 50^2) \\ &= \frac{50}{3}(1600 + 2000 + 2500) \\ &= \frac{50 \times 6100}{3} \\ &= 101{,}666\tfrac{2}{3} \text{ 立方尺} \end{aligned}$$

Method: For a square frustum (方亭):

$$V = \frac{h}{3}(a^2 + ab + b^2)$$

Substituting values:

$$\begin{aligned} V &= \frac{50}{3}(40^2 + 40 \times 50 + 50^2) \\ &= \frac{50}{3}(1600 + 2000 + 2500) \\ &= \frac{50 \times 6100}{3} \\ &= 101{,}666\tfrac{2}{3} \text{ cubic 尺} \end{aligned}$$

6.5 Problem: Round Tower 圓亭臺

圓亭臺

圓亭臺，上周二丈，下周三丈，高二丈。問積幾何？

Round Tower

Problem: A circular platform has top circumference 2 丈, bottom circumference 3 丈, and height 2 丈. What is its volume?

Solution 解:

術曰：上周、下周相乘，又各自乘，并之，以高乘之，三十六而一。

圓臺（圓亭）體積公式：

$$V = \frac{h}{36}(C_1^2 + C_1 \cdot C_2 + C_2^2)$$

其中 C_1、C_2 為上、下周長。中國古代取 $\pi \approx 3$。

Method: For a circular frustum (圓亭), use the formula:

$$V = \frac{h}{36}(C_1^2 + C_1 \cdot C_2 + C_2^2)$$

where C_1 and C_2 are the top and bottom circumferences. In ancient Chinese mathematics, $\pi \approx 3$, so $C = 2\pi r \approx 6r$ and the cross-sectional area $\approx \frac{C^2}{36}$.

6.6 Problem: Grain Pile on the Ground 委粟

委粟

委粟平地，下周一十二丈，高二丈。問積幾何？及為粟各幾何？

Grain Pile on the Ground

Problem: A pile of millet grain on flat ground has base circumference 12 丈 and height 2 丈. What is its volume and how many 斛 of millet does it contain?

Solution 解:

術曰：下周自乘，以高乘之，三十六而一。

穀堆成圓錐形。以 $\pi \approx 3$：

$$\begin{aligned} V &= \frac{C^2 \times h}{36} \\ &= \frac{120^2 \times 20}{36} \\ &= \frac{288{,}000}{36} \\ &= 8{,}000 \text{ 立方尺} \end{aligned}$$

以標準換算率得所容粟數。

Method: The pile forms a cone. Using the formula for a cone with $\pi \approx 3$:

$$\begin{aligned} V &= \frac{C^2 \times h}{36} \\ &= \frac{120^2 \times 20}{36} \\ &= \frac{288{,}000}{36} \\ &= 8{,}000 \text{ cubic 尺} \end{aligned}$$

The conversion to 斛 of grain uses the standard conversion rate.

7 Yang Hui's Triangle 開方作法本源

楊輝最著名之貢獻之一，為保存今稱「楊輝三角」之數表（西方常稱「帕斯卡三角」）。楊輝將此歸功於北宋數學家賈憲（約 1010–1070 年）。

One of Yang Hui's most celebrated contributions is his preservation of what is now known as "Yang Hui's Triangle" (often called Pascal's Triangle in the West). Yang Hui attributes this to the Northern Song mathematician Jia Xian (賈憲, c. 1010–1070).

Yang Hui's Triangle 開方作法本源圖

$$
\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 1 & & 2 & & 1 & & & \\
 & & 1 & & 3 & & 3 & & 1 & & \\
 & 1 & & 4 & & 6 & & 4 & & 1 & \\
1 & & 5 & & 10 & & 10 & & 5 & & 1
\end{array}
$$

開方作法本源：左袤乃積數，右袤乃隅算，中藏者皆廉。以廉乘商方，命實而除之。

此三角提供二項式展開之係數：

$$(a+b)^1 = a+b$$
$$(a+b)^2 = a^2+2ab+b^2$$
$$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$$
$$(a+b)^4 = a^4+4a^3b+6a^2b^2+4ab^3+b^4$$
$$(a+b)^5 = a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$$

Explanation: The left side gives the coefficients for root extraction, the right side gives the corner value (always 1), and the center entries are the "shoulders" (intermediate coefficients). These coefficients are used in the process of extracting roots by the method of increasing multiplication.

The triangle provides the coefficients for binomial expansion:

$$(a+b)^1 = a+b$$
$$(a+b)^2 = a^2+2ab+b^2$$
$$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$$
$$(a+b)^4 = a^4+4a^3b+6a^2b^2+4ab^3+b^4$$
$$(a+b)^5 = a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$$

7.1 Application to Root Extraction 開方應用

楊輝以此係數用於增乘開方法（「遞增乘法」），以求平方根、立方根及更高次方根。此方法等同於西方之霍納法（Horner's method）。

增乘開方法之步驟：

1. 置實（被開方數）
2. 借一算（定位）
3. 以隅法（最高次係數）除實
4. 以次商乘廉法，遞增相乘
5. 逐步逼近，得根之數值

Yang Hui used these coefficients in the method of 增乘開方法 ("Increasing Multiplication Method") for extracting square roots, cube roots, and higher-order roots. This method is equivalent to Horner's method in Western mathematics.

Steps of the Increasing Multiplication Method:

1. Set down the dividend (number to extract root from)
2. Borrow one counting rod for positioning
3. Divide the dividend by the corner method (highest coefficient)
4. Multiply the next quotient by the shoulder method, increasing iteratively
5. Approximate step by step to obtain the numerical root

Appendix: Classification of Nine Chapters Problems 九章纂類

楊輝將《九章算經》二百四十六題，按解法歸納為九大類：

Yang Hui reorganized the 246 problems of the Nine Chapters into nine categories based on their solution methods:

No.	Category 類	Problems
1	乘除 (Multiplication and Division / 乘除)	41
2	分率 (Rates and Proportions / 分率)	41
3	合率 (Combined Rates / 合率)	10
4	互換 (Exchange and Conversion / 互換)	11
5	衰分 (Proportional Distribution / 衰分)	18
6	疊積 (Stacked Accumulations / 疊積)	27
7	盈不足 (Excess and Deficit / 盈不足)	20
8	方程 (Rectangular Arrays / 方程)	19
9	勾股 (Right Triangles / 勾股)	24
	Total 總計	**246**

楊輝竊見九章舊本，作立題法，迫圓古序云：靖康以來，罕有舊本，間有存者，狃於末習，不循本意，以至具術淹廢，偽本滋興。此說固然，殊不知所傳之本亦不得其真矣。

Translation: Yang Hui observed that in old editions of the Nine Chapters, since the Jingkang era (1126 CE), original copies have been rare. Those that survived were corrupted by later practices, deviating from the original intent, leading to the decline of proper methods and the proliferation of erroneous editions.

End of Part II 第二部分終

Bilingual English-Chinese Edition 英漢對照版 — *May 28, 2026*

詳解九章算法

纂類

宋楊輝編次

中英對照本

纂類序

黃帝九章古序云：國家嘗設算科取士，選九章以為算經之首，蓋猶儒者之六經，醫家之難素，兵法之孫子歟。昔聖宋紹興戊辰，算士滎桀謂靖康以來，罕有舊本，間有存者，狃於末習，向獲善本，得其全經，復起於學。以魏景元元年劉徽等、唐朝議大夫、行太史令上輕車都尉李淳風等注釋，聖宋右班直賈憲細草。

輝嘗聞學者，謂九章題問頗隱，法理難明，不得其門而入。於是以答參問，用草考法，因法推類，然後知斯文非古之全經也。將後賢補贅之文，修前代已廢之法，刪立題術，又纂法問，詳著於後，儻得賢者，改而正諸，是所願也。

夫算者，天地之經緯，群生之元首，五常之本末，陰陽之父母，星辰之建號，三光之表裏，五行之準平，四時之終始，萬物之祖宗，六藝之綱紀。稽群倫之聚散，考二氣之降升，推寒暑之迭運，步遠近之殊同，觀天道精微之兆基，察地理縱橫之長短，采神祇之所在，極成敗之符驗。

九章算術，其來尚矣。周公制禮而有九數，九數之流，則九章是矣。昔在庖犧氏始畫八卦，以通神明之德，以類萬物之情，作九九之術，以合六爻之變。暨於黃帝神而化之，引而伸之，於是建曆紀，協律呂，用稽道原，然後兩儀四象精微之氣可得而效焉。

輝不揆淺陋，輒敢刪繁補闕，推類引伸，纂為九類。一曰乘除，二曰互換，三曰合率，四曰分率，五曰衰分，六曰疊積，七曰盈不足，八曰方程，九曰勾股。凡此九類，蓋盡九章之術，而條貫彌綸，秩序井然。使學者由淺入深，由易及難，循序漸進，則九章之法庶幾其可明矣。

楊輝謹序於臨安府学

九章互見目錄

楊輝竊見九章舊本，作立題法遺闕。古序云：靖康以來，罕有舊本，間有存者，狃於末習，不循本意，以至真術淹廢，偽本滋典。此說固然，殊不知所傳之本亦不得其真矣。如粟米章之互換、少廣章之求由開方，皆重疊無謂，而作者題問不歸章次，亦有之。今作纂類，互見目錄，以辯其訛。

古本二百四十六問，其分布如下：

章別	問數		
方田	問		
粟米	問		
衰分	問		
少廣	問		
商功	問		
均輸	問		
盈不足	問		
方程	問		
勾股	問		

纂類九類總目

乘除第一：凡四十一問。方田三十八問，粟米二問，並乘除之術。

互換第二：凡五十六問。粟米三十一問，衰分十一問，均輸十一問，盈不足二問。

合率第三：凡二十問。少廣十一問，均輸八問，盈不足一問。

分率第四：凡九問。本章九問，含乘分、除分、合分、課分、平分諸術。

衰分第五：凡一十八問。本章九問，均輸九問。

疊積第六：凡二十七問。並商功章，含城、垣、堤、溝、渠、壕、倉、窯、盤池、冥谷、芻甍、芻童、羨除等術。

盈不足第七：凡十一問。含盈不足、適足、兩盈、兩不足、盈適足、不足適足之術。

方程第八：凡二十問。本章十八問，均輸盈不足二問，並方程正負之術。

勾股第九：凡三十八問。少廣十三問，商功一問，本章二十四問。

乘除第一

凡四十一問。今考該四十問：方田三十八，粟米二問。此類蓋算學之基礎，凡學算者必先由乘除而入，而後可以進求高深之法。乘除之術不明，則雖遇至易之題，亦將茫然無措矣。

直田法

直田法曰：廣從相乘為實，如畝法而一。按直田即矩形田也，廣者橫之長，從者縱之長，二者相乘得積步，以二百四十步為一畝除之，即得畝數。此術最為簡明，而實諸法之根本。

方田一

廣十五步，從十六步，問田幾何？

答：一畝。

術：廣十五步，從十六步，廣從相乘，得二百四十步，如畝法二百四十步而一，得田一畝。

方田二

廣十二步，從十四步，問田幾何？

答：一百六十八步。

術：廣十二步，從十四步，相乘得一百六十八步。此未滿畝法，故以步計之。

里田法

里田一

方田一里，問為田幾何？

答：三頃七十五畝。

術：方一里自乘，得一里之積；以一里三百七十五畝乘之，即得。

$300 \times 300 = 90{,}000$

圭田法

圭田法曰：半廣以乘正從，或半正從以乘廣。圭田者，三角形田也，其形如圭，故以名之。半廣乘正從者，取三角形之半積也。

圭田一

圭田廣十二步，從二十一步，問田幾何？

答：一百二十六步。

術：半廣得六步，以乘正從二十一步，得一百二十六步。

$6 \times 21 = 126$

邪田法

邪田法曰：併兩斜半之以乘正從，或併兩廣乘半從。邪田者，梯形田也，其兩廣不等，故曰邪。

邪田二

邪田一畔廣三十步，一畔廣四十二步，正從六十四步，問田幾何?

答：九畝一百四十四步。

術：併兩廣，三十步並四十二步，得七十二步，半之得三十六步，以乘正從六十四步，得二千三百四步。如畝法而一，得九畝，餘一百四十四步。

$30+42=7236\times 64=2{,}304$

圓田法

圓田法曰：半周半徑相乘，半周自乘三而一，周自乘十二而一，徑自乘三之四而一。按圓田之術，古法以周三徑一為率，是謂古術。劉徽創割圓之術，以為周三徑一猶為疏闊，乃設密率以究其微。至密率之術，則周自乘，又二十五乘之，三百一十四而一。

圓田一（古術）

圓田周三十步，徑一十步，問田幾何?（古術）

答：七十五步。

術：半周得一十五步，半徑得五步，相乘得七十五步。此用周三徑一之率，半周半徑相乘即得圓積。

$\times$

圓田二（密率）

圓田周三十步，徑一十步，問田幾何?（密率）

答：七十一步二十二分步之一十三。

$71\frac{13}{22}$

術：密率者，周自乘得九百，又二十五乘之，得二萬二千五百，以三百一十四為法除之，得七十一步，餘一百七十六。以法命之，得二十二分步之一十三。

$7171\frac{176}{314}=71\frac{13}{22}$

除率法（商除）

經率法（俗名商除）曰：錢數為實，以所買率為法，實如法而一。原載粟米章。商除者，以商數除實而求其價也，此乃除法之正用。

經率一

錢一百六十文，買瓴甓十八枚，問枚價幾何?

答：一枚八錢九分錢之八。

$8\frac{8}{9}$

術：以錢一百六十文為實，以所買一十八枚為法，實如法而一。不盡者，以法命之，得八錢九分錢之八。

$160\div 18=88\frac{8}{9}$

互換第二

凡五十六問。粟米三十一，衰分十一，均輸十一，盈不足二。互換者，以所有易所求也，凡物之有價者皆可互換，不獨粟米為然。

互換法總論

互換法曰：以所求率乘所有數，為實；以所有率為法，實如法而一。率者，物與物之相為比例者也。知一物之率，則可以推百物；知一時之率，則可以推異時。故算家以率為綱，而互換之術由此出焉。

粟米互換

粟米互換者，以五穀相為變易也。古者粟為政之本，故以粟為主，而以諸米為從。其率之設，則因品質之精粗而異。

粟米互換一

粟二斗一升，問為粺米幾何？

答：一斗一升五十分升之十七。

$\frac{17}{50}$

粟米互換十二

粟一斗，問為糲米幾何？

答：六升。

術：以粟求糲米，三之五而一。粟一斗為十升，以五乘之得五十，如三而一，得一十六升三分升之二。輝按：此答亦與古法不同，當以本術細推之。

$10 \times 5 = 5050 \div 3 = 16\frac{2}{3}$

衰分互換

衰分互換一

今有絲一斤價錢三百四十五，問兩價幾何？

答：一兩二十一錢八分錢之三。

$21\frac{3}{8}$

術：一斤十六兩，以十六兩為法，價錢三百四十五為實，實如法而一。

$345 \div 16 = 2121\frac{9}{16}21\frac{3}{8}$

均輸互換

均輸互換一

今有甲縣粟一石價二十錢，乙縣粟一石價十五錢。甲縣道輸所二百里，乙縣道輸所一百里。問各粟一石至輸所其直幾何？

答：甲縣一石直二十錢，加運價八文，共二十八文；乙縣一石直十五錢，加運價四文，共十九文。

持金出關

持金出關

今持金十二斤出關，稅之十分而取一。今關取金二斤，價錢五千。問金一斤值錢幾何?

答：六千二百五十。

術：以稅金二斤為所有數，價錢五千為所求率，持金十二斤為所有率。以所求率乘所有數，為實；以所有率為法，實如法而一。

$$5000 \times 2 \div 125000 \div (12 \times \frac{1}{10}) \times \frac{1}{\ } = 6250$$

持米出三關

持米出三關，外關三分稅一，中關五分稅一，內關七分稅一，餘存米五斗。問：原米幾何?

答：一十斗九升八分升之三。

術：此題當以還原術推之。餘存米五斗，為內關七分稅一之餘，即六分之五也。以五斗乘七，得三十五斗，以六除之……得一十斗九升八分升之三。

$$\frac{3}{8}$$

$$5 \times 7 \div 6 = 5\frac{5}{6}$$

合率第三

凡二十問：少廣十一，均輸八，盈不足一。合率者，合諸分數而求其率也。其術與分率相為表裏，而所施則異。分率施於乘除，合率施於併合。

少廣法

少廣反合分並率。設諸分母子併而為廣，借田求縱，立少廣章，易合分之術而為之法。用副置分母自乘，以乘全步及諸子，各以本母除其子而併之，免互乘之繁。輝按：少廣之術，其義甚深。廣者所以知從，從者所以知廣。今知廣之不齊，而求從之幾何，是反用合分之術也。

少廣一

田一畝廣一步半，問從？

$1\frac{1}{2}$

答：一百六十步。

術：列置全步及分母子，一步半即二分步之一也。副置分母二自乘，得四。以乘全步一，得四；又以乘子一，得四。各以本母除子，併之為法。輝按：此術以分母自乘通之，法得三，實得四百八十，實如法而一，得一百六十步。

$1\frac{1}{2} = \frac{3}{2}480 \div 3 = 160$

少廣二

田一畝廣一步半，三分步之一，問從？

$1\frac{1}{2} + \frac{1}{3}$

答：一百三十步一十一分步之一十。

$130\frac{10}{11}$

術：列置全步及分母子。副置分母二、三，相乘得六為通分。以六乘全步一，得六；以六乘二分之一子一，得六，以母二除之得三；以六乘三分之一子一，得六，以母三除之得二。併之得十一為法。以畝法二百四十步乘通分六，得一千四百四十為實。實如法而一，得一百三十步，餘十。以法命之。

$6 \times 1 = 66 \times \frac{1}{2} = 36 \times \frac{1}{3} = 26 + 3 + 2 = 11240 \times 6 = 1440$

$1440 \div 11 = 130130\frac{10}{11}$

少廣十一

田一畝廣一步半，三分步之一，四分步之一，五分步之一，六分步之一，七分步之一，八分步之一，九分步之一，十分步之一，十一分步之一，十二分步之一，問從？

$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12}$

$77\frac{86984}{45813}$

$\frac{83711}{27720}$

答：七十七步四萬五千八百一十三分步之八萬六千九百八十四。

分率第四

凡九問。分率者，分數之率也。算術之難，莫難於分數。分數之難，在於分子分母之錯綜交互，非精通其理者不能得其正。九章設分率一章，專論分數之四則運算，以為諸術之基礎。

乘分法

乘分法曰：分母各乘其全，分子從之，相乘為實，分母相乘為法。乘分者，分數相乘之術也。其理以分母通全，而後相乘，所以齊其不齊也。

乘分一

田廣一十八步七分步之五，從二十三步十一分步之六，問為田幾何？

$18\frac{5}{7}23\frac{6}{11}$

答：一畝二百步十一分步之七。

$\frac{7}{11}$

術：廣一十八步七分步之五，分母七乘全步一十八，得一百二十六，併分子五，得一百三十一為廣實。從二十三步十一分步之六，分母十一乘全步二十三，得二百五十三，併分子六，得二百五十九為從實。以廣實乘從實，得三萬三千九百二十九為積實。分母七乘十一，得七十七為法。

$7\times18+5=13111\times23+6=259131\times259=33{,}9297\times11=77$
$33{,}929\div77=440\frac{52}{77}=\frac{7}{11}$

除分法

除分一

七人均入錢三分錢之一，問人得幾何？

$1\frac{1}{3}$

答：人得一錢二十一分錢之四。

$1\frac{4}{21}$

術：以人數七為法，錢三分錢之一通之，得三分之四為實。實如法而一，人得二十一分之四錢。

$1\frac{1}{3}=\frac{4}{3}\frac{4}{3}\div7=\frac{4}{21}\frac{4}{21}1\frac{4}{21}$

合分法

合分法曰：母互乘子，併以為實，母相乘為法。合分者，併眾分數為一也，今謂之加法。

合分一

三分之一，五分之二，問合之得幾何？

$\frac{1}{3}+\frac{2}{5}$

答：一十五分之十一。

$\frac{11}{15}$

術：母互乘子，三乘二得六，五乘一得五，併之得十一為實。母相乘，三乘五得十五為法。

$3\times2=65\times1=56+5=113\times5=15$

合分二

三分之二，四分之三，五分之四，問合之得幾何？

$\frac{2}{3}+\frac{3}{4}+\frac{4}{5}$

答：一六十分之四十三。

$\frac{49}{60}\frac{43}{60}$

術：三母連乘得六十為法。三乘三乘四得三十六為第一子；四乘二乘四得三十二為第二子；五乘二乘三得三十為第三子。併之得九十八為實。實如法而一，得一六十分之九十八，約為四十九分之五十二。

$3\times4\times5=604\times5\times2=403\times5\times3=453\times4\times4=48$
$133/60=2\frac{13}{60}$

課分法

課分一

三分之二，四分之三，問孰多？多幾何？ $\frac{2}{3}\frac{3}{4}$

答：四分之三多，多十二分之一。 $\frac{3}{4}\frac{1}{12}$

術：母互乘子，三乘三得九，四乘二得八，以少減多，九減八餘一為實。母相乘，三乘四得十二為法。 $3 \times 3 = 94 \times 2 = 83 \times 4 = 12\frac{1}{12}$

平分法

平分一

三分之二人出二，四分之三人出三，問人各應出幾何？

答：各出二錢十三分錢之六。 $2\frac{6}{13}$

衰分第五

凡一十八問：本章九問，均輸九問。衰分者，差分之法也。以差率為比例而分配之，使各得其當。衰有列衰、返衰之別，列衰者，以多為多，以少為少；返衰者，以多為少，以少為多。

衰分法曰：各置列衰，副併為法，以所分乘未併者各自為實，實如法而一。輝按：衰分之術，施於賦役之征調，祿食之頒給，財貨之分配，無所不宜。

衰分一（牛羊豕價）

今有牛二羊五買豕一十三家，有餘錢一千；賣牛三豕三以買九羊，錢適足；賣六羊八豕以買五牛，錢不足六百。問牛羊豕價各幾何？

$2O + 5S - 13P = 10003O - 9S + 3P = 05O - 6S - 8P = -600$

答：牛價一千二百，羊價五百，豕價三百。

術：此題當以方程術解之。設牛價為牛，羊價為羊，豕價為豕。牛二加羊五減豕一十三，餘一千；牛三減羊九加豕三，適足；牛五減羊六減豕八，不足六百。列為三行，以正負術入之。

衰分二（五雀六燕）

五雀六燕共重一斤，雀重燕輕，交易一枚，其重適等。問各幾何？

$1\frac{13}{19}\frac{5}{19}$

$= 5S + 6 \times \frac{3}{4}S = 5S + 4.5S = 9.5S = 16S = \frac{32}{19} = 1\frac{13}{19}$

答：雀一兩十九分兩之一十三，燕一兩十九分兩之五。

術：五雀六燕共重一斤，即十六兩。交易一枚其重適等者，四雀一燕之重與五燕一雀之重適等，即三雀與四燕等重。以三雀當四燕，則五雀六燕共為九雀又三分之二雀，以十六兩除之，得雀之重。又以三雀四燕之比例，得燕之重。

衰分三（返衰）

今有大夫五人，不更四人，簪裊三人，上造二人，公士一人，凡十五人，共出百五十錢。欲令爵高者出少，以次漸多，問各幾何？

$5(1) + 4(2) + 3(3) + 2(4) + 1(5) = 5 + 8 + 9 + 8 + 5 = 35$
$150 \times 1/35 \times 5 = \ldots$

答：大夫五人各出十錢，不更四人各出十五錢，簪裊三人各出二十錢，上造二人各出二十五錢，公士一人出五十錢。

術：此返衰之術也。爵高者出少，以次漸多，故用返衰。大夫五人以一為衰，不更四人以二為衰，簪裊三人以三為衰，上造二人以四為衰，公士一人以五為衰。

衰分六（女子善織）

今有女子善織，日自倍，五日織五尺。問日織幾何？

$1\frac{19}{31} 3\frac{7}{31} 6\frac{14}{31} 12\frac{28}{31} 25\frac{25}{31}$

$50 \times 1/31 = 1\frac{19}{31} 50 \times 2/31 = 3\frac{7}{31}$

答：初日織一寸三十一分寸之十九，次日織三寸三十一分寸之七，第三日織六寸三十一分寸之十四，第四日織一尺二寸三十一分寸之二十八，第五日織二尺五寸三十一分寸之二十五。

術：此題以倍衰為列衰。初日為一，次日為二，第三日為四，第四日為八，第五日為十六。併之得三十一為法。以五尺乘未併者各自為實，實如法而一。

疊積第六

凡二十七問，並商功章。疊積者，積疊之術也。凡築城、穿壕、開渠、建仓，皆需度土方之多寡，此疊積之所由設也。商功之術，以立方為本。立方者，長廣高深相乘之積也。

$\times\times$

城垣壕溝

城垣一

城下廣四丈，上廣二丈，高五丈，袤一百二十六丈五尺。問積幾何?

答：一百八十九萬七千五百尺。

術：併上下廣，得六丈，半之得三丈，以高五丈乘之，得十五萬平方尺，以袤一百二十六丈五尺乘之，即得積尺。

$(4+2)/2=33\times 5=1515\times 126.5=1897.5$= $1{,}897{,}500^{33}$

倉窖積粟

倉一

方倉廣一丈六尺，長二丈，深三尺。問積及容粟幾何?

答：積九百六十尺，容粟九百六十斛。

術：方倉之積，廣乘長，又乘深，即得積尺。以二尺七寸為一斛之積除之，即得容粟之數。輝按：古者量制，一斛之積為二尺七寸，故以積尺除之得斛數。

$16\times 20\times 3=960960\div 1=960$

芻甍芻童

芻甍

芻甍法曰：倍下長併入上長，以廣乘之，又高乘之，皆如六而一。下廣三丈，袤四丈，上袤二丈，無廣，高一丈。問積幾何?

答：五千尺。

術：倍下長得八丈，併入上長二丈，得十丈。以下廣三丈乘之，得三十萬平方尺。又以高一丈乘之，得三百萬立方尺。如六而一，得五千尺。

$2\times 4=88+2=10$= $30{,}000\div 6=5{,}000$

芻童

芻童法曰：倍上長併入下長，以下廣乘之；又倍下長併入上長，以上廣乘之；併之，以高乘之，皆如六而一。上廣二丈，上袤三丈，下廣四丈，下袤五丈，高二丈。問積幾何?

答：二萬六千尺。

$\times\times\times\div\times\times$

$(2\times 3+5)\times 4+(2\times 5+3)\times 2=11\times 4+13\times 2=44+26=70$
$\times 2=140$= $140{,}000^{3}\div 6=23{,}333\frac{1}{3}$

陽馬

陽馬法曰：廣袤相乘，以高乘之，三而一。廣五尺，袤七尺，高八尺。問積幾何？

$\times\times\div$

答：九十三尺三分尺之一。

$93\frac{1}{3}$

術：廣五尺乘袤七尺，得三十五平方尺。又以高八尺乘之，得二百八十立方尺。三而一，得九十三尺三分尺之一。

$5\times 7 = 3535\times 8 = 280280\div 3 = 93\frac{1}{3}$

鱉臑

鱉臑法曰：廣袤相乘，又以高乘之，六而一。廣四尺，袤五尺，高三尺。問積幾何？

$\times\times\div$

答：一十尺。

術：廣四尺乘袤五尺，得二十平方尺。又以高三尺乘之，得六十立方尺。六而一，得一十尺。

$4\times 5 = 2020\times 3 = 6060\div 6 = 10$

盈不足第七

凡十一問。盈不足者，假設之術也。凡數之隱晦難知者，設兩假以測之，一假則盈，一假則不足，因盈不足以定其實。此術之妙，在於以虛測實，以假求真。

盈朒適足法

盈朒適足法曰：置所出率，盈朒各居其下。副置出率，以少減多餘為約法。盈朒適足，令維乘所出率實，以盈朒之數乘人積為人實，實皆如約法而一。輝按：盈不足之術，源遠流長。其法以兩次假設之結果，一盈一不足，由此推求人數與物價。其理與今之線性插值法相通。

盈不足一（共買犬）

共買犬，人出五不足九十文，人出五十文適足。問人數、犬價各幾何？

答：二人，犬價一百。

術：置所出率，五文與五十文。以少減多，五十減五餘四十五為法。以盈不足之數，不足九十為實。實如法而一，得二人，即人數。以適足出率五十乘人數二，得一百，即犬價。驗之：人出五，二人共出十，不足九十，則犬價為一百；人出五十，二人共出一百，適足。合問。

$50 - 5 = 4590 \div 45 = 250 \times 2 = 1005 \times 2 = 1050 \times 2 = 100$

盈不足三（買瓦）

共買瓦，人出八盈三枚，人出七不足四枚。問人數、瓦數各幾何？

答：七人，瓦五十三枚。

術：置所出率八與七，以少減多餘一為法。盈三不足四，併之得七為實。實如法而一，得七人。以盈三減之，或以不足四加之，皆得瓦五十三枚。

$3 + 4 = 77 \div 1 = 78 \times 7 - 3 = 537 \times 7 + 4 = 53$

盈不足六（二酒求價）

今有醇酒一斗直錢五十，行酒一斗直錢一十。今將錢三十，得酒二斗。問醇行酒各幾何？

答：醇酒二升半，行酒一斗七升半。

術：此題以盈不足術解之。設醇酒一斗，則直錢五十，已過三十；設行酒一斗，則直錢一十，不足二十。以盈不足術求之，醇酒得二升半，行酒得一斗七升半。輝按：此術即今之所謂雞兔同籠也。

$2\frac{1}{2}$

$= \frac{30-10}{50-10} = \frac{20}{40} = \frac{1}{4} = \frac{1}{2} = 52.5$

盈不足七（五家共井）

今有五家共井，甲二綆不足如乙一，乙三綆不足如丙一，丙四綆不足如丁一，丁五綆不足如戊一，戊六綆不足如甲一。如各得所不足一綆，皆逮。問井深、綆長各幾何？

$a, b, c, d, e D 2a + b = D 3b + c = D 4c + d = D 5d + e = D$
$6e + a = D$

方程第八

凡二十問：本章十八問，均輸盈不足二。方程者，方陣之程術也。以行列相併相減，求眾物之價值，此最為精妙之術也。輝按：方程之術，乃中國數學之瑰寶也。其法以算籌列為方陣，互乘相減，以消去未知數，此與今之代數學所謂行列式、矩陣者異曲同工。

方程法

方程法曰：所求率互乘鄰行，以少減多，再求減損。錢為實，物為法，實如法而一。置位法曰：依所問排列逐物與價，而鄰行相對如之。

方程一（上中下禾）

上禾三束，中禾二束，下禾一束，共實三十九斗。上禾二束，中禾三束，下禾一束，共實三十四斗。上禾一束，中禾二束，下禾三束，共實二十六斗。問：上、中、下禾一束各幾何？

答：上禾一束，得九斗四分斗之一；中禾一束，得四斗四分斗之一；下禾一束，得二斗四分斗之三。

$9\frac{1}{4}4\frac{1}{4}2\frac{3}{4}$

$3u+2m+l=392u+3m+l=34u+2m+3l=26lu-m=5$

方程二（牛羊價）

五牛二羊，直金十兩；二牛五羊，直金八兩。問：牛、羊價各幾何？

答：一牛直金一兩二十一分兩之一十三；一羊直金二十一分兩之二十。

$1\frac{13}{21}\frac{20}{21}$

$5O+2S=102O+5S=810O+4S=2010O+25S=40$
$21S=20S=\frac{20}{21}5O=10-\frac{40}{21}=\frac{170}{21}O=\frac{34}{21}=1\frac{13}{21}$

正負術

正負術曰：同名相除，異名相益，正無入負之，負無入正之。其異名相除，同名相益，正無入正之，負無入負之。輝按：正負之術，乃方程之樞紐也。古者以赤籌為正，黑籌為負，列於算板之上，以相消長。同名相除者，正與正除，負與負除也；異名相益者，正與負益也。

方程四（賣牛買羊）

今有賣牛二、羊五，以買一十三豕，有餘錢一千；賣牛三、豕三，以買九羊，錢適足；賣六羊、八豕，以買五牛，錢不足六百。問：牛、羊、豕價各幾何？

答：牛價一千二百，羊價五百，豕價三百。

$2O+5S-13P=10003O-9S+3P=05O-6S-8P=-600$

術：此三行方程也。設牛價為正，羊價為負，豕價為負，餘錢為正，不足為負。列為三行，以正負術入之。

方程六（麻麥豆）

今有麻三斗、麥四斗、豆五斗，共直錢一百；麻四斗、麥五斗、豆三斗，共直錢一百一十；麻五斗、麥三斗、豆四斗，共直錢一百二十。問：麻、麥、豆一斗各直幾何？

$H+W-2B=10H=15W=5B=8$

方程七（五色絲）

今有青絲五斤、黃絲四斤、白絲三斤、朱絲二斤、黑絲一斤，共直錢一百；青絲四斤、黃絲三斤、白絲二斤、朱絲一斤、黑絲五斤，共直錢九十；青絲三斤、黃絲二斤、白絲一斤、朱絲五斤、黑絲四斤，共直錢八十；青絲二斤、黃絲一斤、白絲五斤、朱絲四斤、黑絲三斤，共直錢七十；青絲一斤、黃絲五斤、白絲四斤、朱絲三斤、黑絲二斤，共直錢六十。問：五色絲一斤各直幾何？

勾股第九

凡三十八問：少廣十三，商功一問，本章二十四。勾股者，直角三角形之術也。勾為短邊，股為長邊，弦為斜邊。勾股之術，以自乘併之為根本，以開方為施用。此術之用，至為廣大，凡測高深、量遠近、求曲直，無不用之。

勾股基本方法

勾股法曰：勾自乘，股自乘，併之為弦實，開方除之即弦。輝按：勾股之術，其源甚古。周公問於商高曰：「數之法出於圓方，圓出於方，方出於矩，矩出於九九八十一。故折矩以為勾廣三，股修四，徑隅五。」此勾股之術所由起也。

勾股一（立木垂索）

立木垂索，委地二尺，引索斜之，拄地去木八尺。問索長幾何？

答：一十七尺。

術：此題以勾股術解之。立木垂索，委地二尺，此股弦較也。拄地去木八尺，此勾也。勾八尺自乘，得六十四。如股弦較二尺而一，得三十二。加較二尺半之得一……輝按：此術當以勾股定理細推之。勾八尺，股十五尺，弦一十七尺。八平方加十五平方，得六十四加二百二十五，得二百八十九，開方得一十七。合問。

$xian - gu = 28^2 + gu^2 = xian^2 8^2 = xian^2 - gu^2 = (xian - gu)(xian+gu) = 2(xian+gu) = 64 xian+gu = 32 xian-gu = 22 \cdot xian = 34 xian = 17 gu = 158^2 + 15^2 = 64 + 225 = 289 = 17^2$

勾股三（圓材鋸深）

圓材埋在壁中，不知大小，鋸深一寸，道長一尺。問徑幾何？

$r^2 = (r -)^2 + (/2)^2 r^2 = (r - 1)^2 + 25 = r^2 - 2r + 1 + 25\ 2r = 26 r = 13$

勾股容方容圓

勾股容方

今有勾五步，股十二步。問：勾中容方幾何？

答：方三步十七分步之九。

$3\frac{9}{17}$

術：勾股容方之術，以勾股相乘為實，併勾股為法，實如法而一，即得方之邊。勾五步乘股十二步，得六十為實。併勾股得十七為法。實如法而一，得三步十七分步之九。輝按：容方之術，以勾股為邊之直角三角形內作正方形，其一邊在勾上，一邊在股上，一頂點在弦上。

$\frac{\times}{+} = \frac{5\times12}{5+12} = \frac{60}{17} = 3\frac{9}{17}$

勾股容圓

今有勾八步，股十五步。問：勾股中容圓徑幾何？

答：六步。

術：勾股容圓之術，以勾股相乘倍之為實，以勾股弦併之為法，實如法而一，即得圓徑。勾八步乘股十五步，得一百二十，倍之得二百四十為實。勾八步、股十五步、弦十七步併之，得四十為法。實如法而一，得六步，即圓徑。輝按：此術之證，可以面積法求之。勾股三角形之面積，等於其周界與圓徑乘積之四分之一。

$d = \frac{2\times\times}{++} = \frac{2\times8\times15}{8+15+17} = \frac{240}{40} = 6\frac{1}{2}\times\times\frac{1}{4}\times\times d$

複雜勾股問題

勾股六（竹折）

今有竹高一丈，末折抵地，去本三尺。問：折者高幾何？

答：四尺五十五分尺之十一。

$4\frac{11}{55} = 4\frac{1}{5}$

術：此題以勾股術解之。竹高一丈為股弦之和，去本三尺為勾。以勾股術入之，即得折處之高。

$h10 - h3^2 + h^2 = (10 - h)^2 = 100 - 20h + h^2 9 = 100 - 20h$
$h = \frac{91}{20} = 4\frac{11}{20}$

勾股七（葭生中央）

今有池方一丈，葭生其中央，出水一尺。引葭赴岸，適與岸齊。問：水深、葭長各幾何？

術：此題以勾股術解之。池方一丈，葭在中央，赴岸為五尺，此勾也。水深為股，葭長為弦。出水一尺為股弦較。輝按：此題為勾股章中最有名之題，歷代算家多稱道之。以葭之出水，而知池之深，此可謂以近知遠，以顯知幽者也。

$-5^2 = (xian - gu)(xian + gu) = 1 \times (xian + gu) xian + gu = 25$
$xian - gu = 12 \cdot xian = 26$

勾股八（邑方）

今有邑方不知大小，各開中門。出北門二十步有木，出南門十四步折而西行一千七百七十五步見木。問：邑方幾何？

$s\frac{s/2+20}{1775} = \frac{s/2}{14+s/2} s = 250$

勾股九（山高海遠）

今有望海島，立兩表，高三丈，前後相去千步。令後表與前表參相直。從前表卻行一百二十三步，人目著地，取望島峯，與表末參合。從後表卻行一百二十七步，人目著地，取望島峯，亦與表末參合。問：島高及去表各幾何？

$= \frac{\times}{} + = \frac{3\times1000}{127-123} + 3 = 750 + 3 = 753 = \frac{123\times1000}{4} = 30{,}750$

術：此題以重差術解之，為勾股術之推廣。以兩表之高及相去之遠，與兩卻行之差，以比例求之，即得島高及去表之遠。輝按：此題為劉徽海島算經之第一題，雖不在九章之內，然其術出於勾股。

附錄：纂類九類總論

楊輝纂類，將九章二百四十六問重新分為九類，此實為中國數學史上的一大創舉。古本九章以方田、粟米、衰分、少廣、商功、均輸、盈不足、方程、勾股分章，此以數學之應用領域為分類標準，便於官府之施用，而不利於學者之研習。輝有感於此，乃以數學之方法為分類標準，重新編排，使學者可以由淺入深，循序漸進。

輝之九類：一曰乘除，二曰互換，三曰合率，四曰分率，五曰衰分，六曰疊積，七曰盈不足，八曰方程，九曰勾股。此九類之中，乘除、分率為基礎之術，互換、衰分為比例之術，合率為分數之併合，疊積為體積之計算，盈不足為假設之法，方程為聯立方程之解法，勾股為幾何之基礎。由淺入深，由易及難，條理井然。

乘除之類

乘除者，算學之基礎也。一切算術，無不始於乘除。楊輝以方田三十八問、粟米二問列入此類，蓋方田之計算，無非廣從相乘，以得積步；而粟米之經率，亦以除法為主。此類之術雖簡，然為諸法之根本，不可不精熟也。

互換之類

互換之術，出於粟米章，而其用則遍於衰分、均輸諸章。以所有易所求，以所求率乘所有數為實，以所有率為法，實如法而一。此術之要，在於明率。率者，物與物之相為比例者也。知一物以推百物，知一時以推異時，此互換之妙用也。

方程之類

方程之術，乃中國數學之瑰寶也。以算籌列為方陣，互乘相減，以消去未知數。正負之術，為其樞紐。同名相除，異名相益，正無入負之，負無入正之。此術之妙，非深思不能得其旨也。

勾股之類

勾股之術，以自乘併之為根本，以開方為施用。測高深、量遠近、求曲直，無不用之。容方、容圓之術，尤為精妙。重差之術，由是而推廣，可以測日之高遠、海島之廣狹，此其大用也。

後記

輝不自揣，以淺陋之學，妄議古人之書。然區區之意，欲使九章之法，燦然大明於後世，使學者無復有隱晦難明之歎。纂類之作，非敢妄改舊章，不過條分縷析，使條貫更清晰耳。昔劉徽注九章，自謂「幼習九章，長再詳覽，觀陰陽之割裂，總算術之根源，探賾之暇，遂悟其意。是以敢竭頑魯，采其所見，為之作注。」輝雖不敢比擬劉徽，然其用心則一也。

九章算術，為中國數學之源泉，兩千年來，歷代算家無不由此入手。輝之詳解，不過為初學者設階梯耳。至於深造自得，則在乎人之力矣。

楊輝識

楊輝（約），字謙光，錢塘（今浙江杭州）人，南宋傑出數學家。著有《詳解九章算法》十二卷、《日用算法》二卷、《乘除通變本末》三卷、《田畝比類乘除捷法》二卷、《續古摘奇算法》二卷，後三種合稱《楊輝算法》。楊輝三角之發現，尤為世人所稱道，實開二項式係數研究之先聲。

第三部終